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1. Introduction

This coursework is all about the report on the structured analysis of the Gokyo Bistro Online Management System. Gokyo Bistro is a popular restaurant located in Jhamsikhel, Lalitpur and it is facing some operational challenges due to increasing customer demand, inefficient table management and growing home delivery requests.

Our system aims to automate restaurant operations such as table reservation, home delivery ordering, billing, reporting and feedback management. The analysis has been carried out using the Structured Software Engineering, that includes Data Flow Diagrams (DFDs), Entity Relationship Diagram (ERD), Data Dictionary and process specifications. For this we are working on a group as a team and our team is of total five members. Each and every member of the group is equally involved and has contributed to produce the best out of this project. The teamwork and the different perspectives of individual members lead us to solve the problems in efficient and best ways possible.

We properly studied the problems of the Gokyo Bistro restaurant and figured out the solution majors by the help of different entities and process that directly or indirectly affects the operations of the restaurant. With the proper implementation of our project as a solution we are pretty sure and confident about overcoming the challenges that the restaurant is facing.

2. Business Case

2.1. Problem statement

Gokyo Bistro usually manages table reservations and orders manually. The customer flow is high and due to this, the restaurant is facing the problems mentioned below:

- Customers leave because of full occupancy
- Table availability is poorly managed
- Home delivery requests are increasing
- Financial reporting is not predictive

2.2. Business rule/requirements

The business rule or the requirements of the coursework specifications are described below:

- **User Hierarchy:** The system recognizes three user tiers, Admin (management), Member (registered customer) and Visitor (guest users).
- **Booking Financials**: A table reservation is not confirmed until the user pays a mandatory advance of Rs. 1200.
- **Standard Refund Policy:** If a reservation is cancelled before the scheduled date, the system must process a 100% refund of the advance.
- **Late Cancellation Policy**: If a reservation is cancelled on the same day of the booking, the system must deduct a penalty fee of Rs. 500 and refund only the remaining amount.
- **Exclusive Delivery:** The “Home Delivery” feature is restricted strictly to Members. Visitors are not authorized to use this feature.
- **Beverage Privileges:** While visitors can select general beverage preferences. Only members are permitted to especially reserve wine or whiskey bottles.
- **Administrative Control:** Only the Admin has the authority to update the menu, post discount offers, and view financial forecasts.

2.3. Proposed solution

The proposed solution to eradicate the problems and meet the requirements are as follows and the system:

- Allows advance table booking with payment
- Supports member-based home delivery
- Automates billing and receipt generation
- Provides financial reports and revenue prediction
- Collects customer feedback

2.4. Financial Analysis (Assumed)

2.4.1. Financial assumptions

- Average table reservations per day: 20
- Advance payment for reservation: NPR 1200
- Same-day cancellation rate: 10%
- Cancellation deduction amount: NPR 500
- Home delivery orders per day (members): 12
- Average home delivery order value: NPR 1,400
- Operating days per month: 30

2.4.2. Revenue from each section

Table: Monthly reservation count: 20 * 30 = 600 reservations

Monthly Advance revenue: 600 * 1200 = NPR 720,000

Cancellation Deductions: Number of same-day cancellations: 10% of 600 = 60 cancellations

Monthly cancellation revenue: 60 * 500 = NPR 30,000

Home delivery orders: Monthly home delivery orders: 12 * 30 360 orders

Monthly home delivery revenue: 360 * 1400 = NPR 504,000

2.4.3. Total estimated monthly and annual revenue

Table 1 monthly and annual revenue estimation

Revenue source	Amount (NPR)
Table reservation advance	720,000
Cancellation deduction	30,000
Dome delivery orders	504,000
Total Monthly revenue	1,254,000
Total annual revenue	15,048,000

2.4.4. Estimated operational costs

Table 2 estimated operational costs

Cost category	Monthly Cost (NPR)
Server hosting and maintenance	40,000
Payment gateway charges	25,000
System maintenance	20,000
Miscellaneous costs	15,000
Total monthly cost	100,000

2.4.5. Net Monthly and Annual benefit

Net monthly benefit: 1,254,000 – 100,000 = NPR 1,154,000

Net annual benefit: 1,1154,000 * 12 = NPR 13,848, 000

2.4.6. Financial forecast and growth

With improved customer experience and efficient management:

- Expected annual growth: 10-15%
- Increased repeat customers through membership
- Better menu optimization using sales data

2.4.7. Business Benefits

After all the changes and upgrades the system will have:

- Reduced customer waiting time
- Increased revenue through better planning
- Improved customer satisfaction
- Better management decision-making

2.5. Software Requirements Specifications (SRS)

A Software Requirements Specifications (SRS) is a detailed document that explains the purpose, scope and operating environment of a software system being developed. It clearly describes what the software should do and how it is expected to perform. As SRS helps reduce development time and cost by giving developers a clear understanding of requirements. It also explains how the software will interact with hardware, other software and users in real situations. Key factors such as performance, response time, reliability, security, maintainability and recovery from failures are defined in the SRS (Nick Barney, Linda Rosencrance, 2024).

The different Software Requirements Specifications of the project are as follows:

2.6. Functional Requirements

All the functional requirements are listed below:

1. **Authentication:** The system shall allow Admins and Members to log in securely using unique credentials.
2. **Table Reservation:** The system shall allow visitors and members to check table availability and book a slot. The system shall only change status to “Confirmed” upon successful receipt of the Rs. 1200 advance.
3. **Cancellation Logic:** The system shall automatically compare the cancellation date with the booking date to determine if the refund should be full or if the Rs. 500 penalty applies.
4. **Home Delivery Module:** The system shall validate the user type, if the user is a Member, the system shall allow food ordering and online payment processing.

5. **Reporting:** The system shall generate financial reports and utilize historical data to display a revenue prediction for future years.
6. **Billing:** The system shall generate detailed internal bills for the admin and printable receipts for the customer.
7. **Feedback:** The system shall allow users to rate Food, Staff and Ambience on a scale and provide text comments.

2.7. Non-Functional Requirements

The non-functional requirements include:

1. **Security:** All payment transactions must be encrypted. Admin access must be secured to prevent unauthorized menu or price changes.
2. **Performance:** The system must handle simultaneous booking requests without data conflict.
3. **Availability:** The web application must be available 24/7 to maximize booking potential.
4. **Usability:** The interface must be responsive and intuitive, requiting no training for visitors to use.

2.8. Design and Implementation Constraints

The design and implements constraints consist of:

1. **Web-based Architecture:** The system must be developed as a web application accessible via standard browsers to ensure accessibility for all users without app installation.
2. **Currency Standardization:** All financial calculations must be processed in NPR (Nepalese Rupee).
3. **Internet Dependency:** The system requires an active internet connection to check live table inventory; offline mode is not required.

2.9. External Interface Requirements

The external interface requirements include hardware and software requirements and they are:

- **User Interfaces:** The system shall feature a responsive web interface that adapts to desktop and mobile devices.
- **Hardware Interfaces:**
  1. **Thermal Printer:** The system must interface with a standard restaurant printer to generate physical receipts.
  2. **Server:** The system requires a hosting server to manage the centralized database.
- **Software Interfaces:**
  1. **Payment Gateway API:** The system must integrate with an external provider like bank, Khalti, eSewa to process the Rs. 1200 advance and delivery payments.
  2. **Database Management System:** To store persistent data for users, reservations and orders.

2.10. Environmental model specification

2.10.1. Context level diagram

It is a general overview of a system and is usually shown using a high-level DFD called a context diagram of Level 0 DFD. It provides a simple and clear view of the system, making it easy for a wide audience to understand. This diagram includes only one process, known as process 0, which represents the entire system and its overall functionality (Orakzai, 2023). The DFD level 0 of our project is presented below:

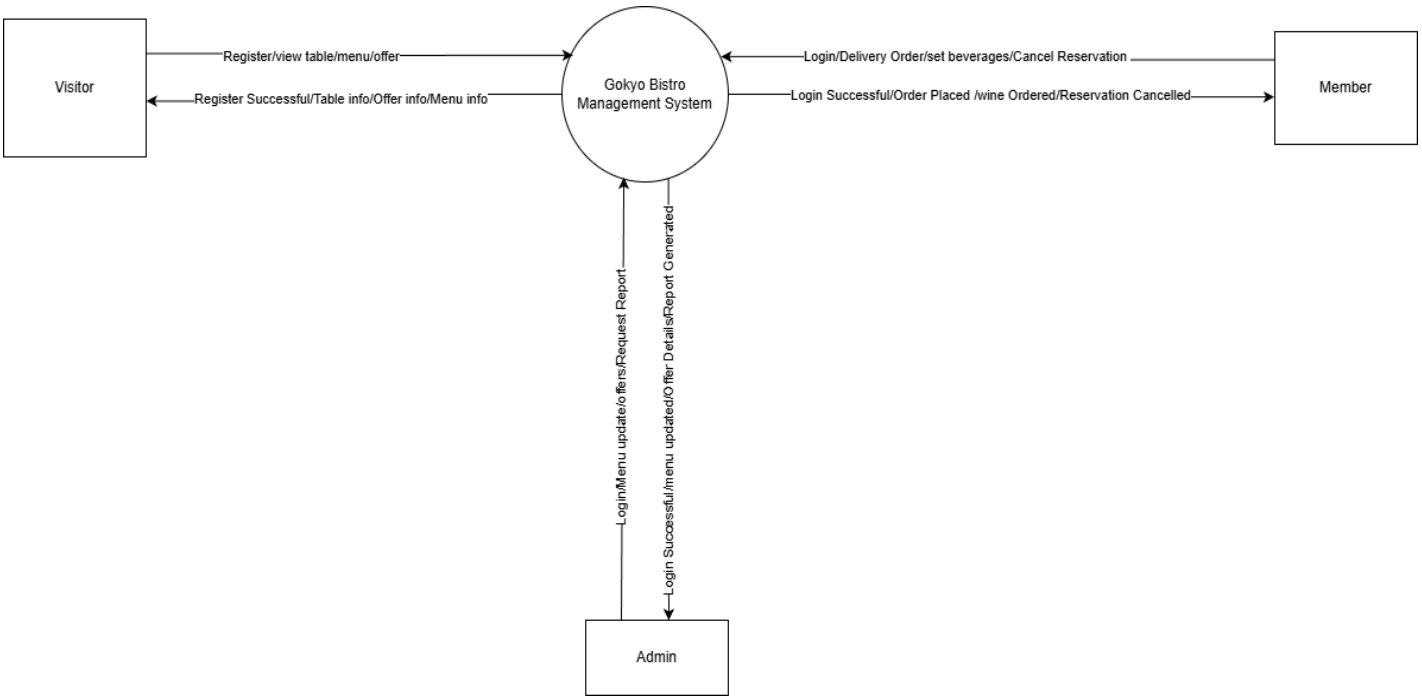


Figure 1 Level 0 DFD

2.11. Internal model specification of the system

A data flow diagram (DFD) is a visual representation that shows how data flows within a system. It displays data inputs, outputs, process and data stores, helping to explain how data is handled and transformed. DFDs give a clear, high-level view of how a system works and are easy to understand for both technical and non-technical users. They are commonly used to show the main processes and how information moves between different parts of a system (Geeksforgeeks, 2025).

2.11.1. DFD level 1

Level 1 DFD give a broader view of a system but with more detail than level 0 diagram. They expand the single process shown in the context diagram into multiple subprocesses, making it easier to understand the system's main functions and how data flows between them (Orakzai, 2023). The DFD level 1 of our project is presented below:

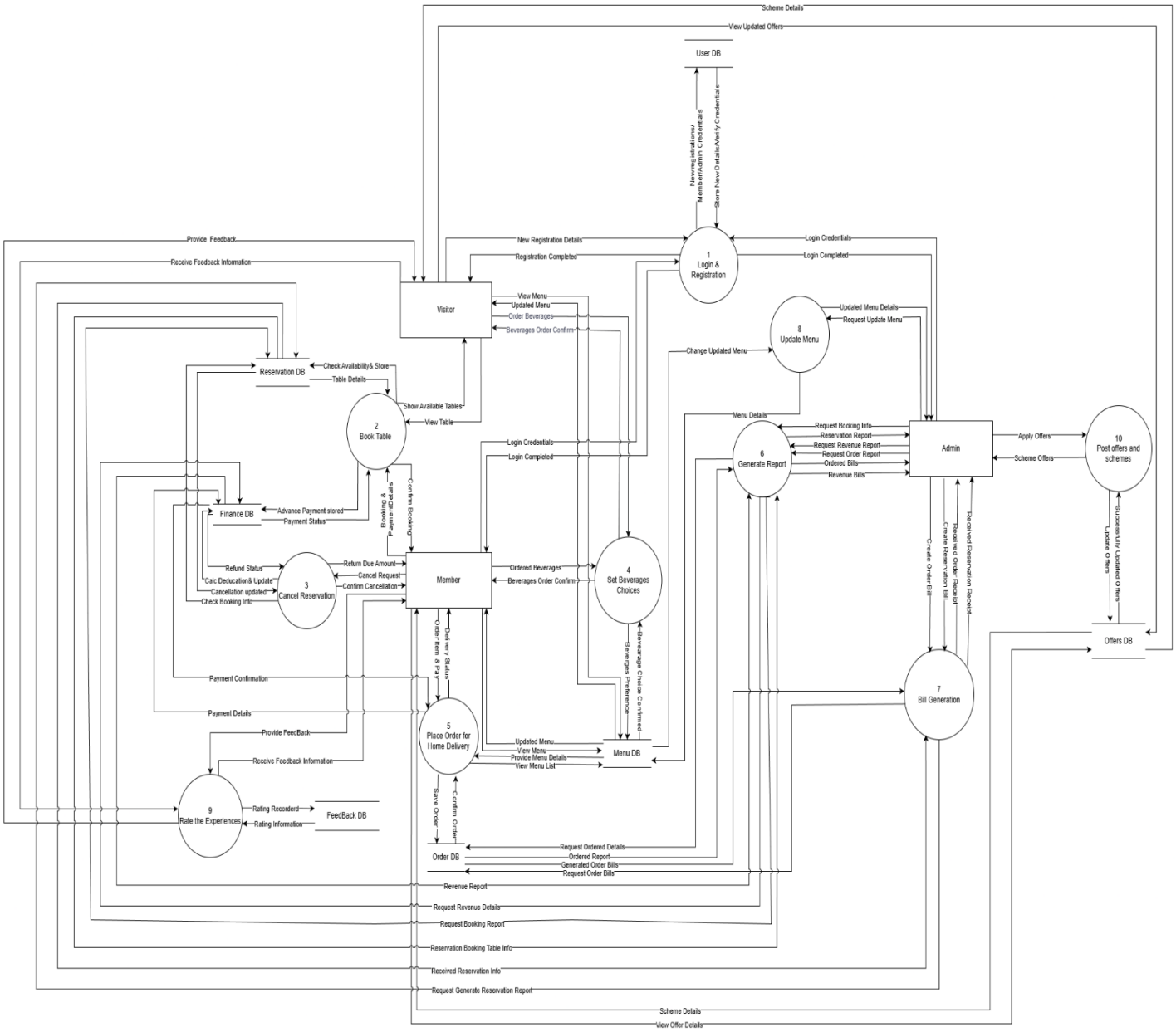


Figure 2 Level 1 DFD

2.11.2. DFD level 2

The next level of DFDs adds more detail by further breaking each level 1 process into smaller subprocesses. This deeper view helps in clearly understanding how the system works and make it easier to spot inefficiencies or areas that can be improved (Orakzai, 2023). The DFDs level 2 of our project are presented below:

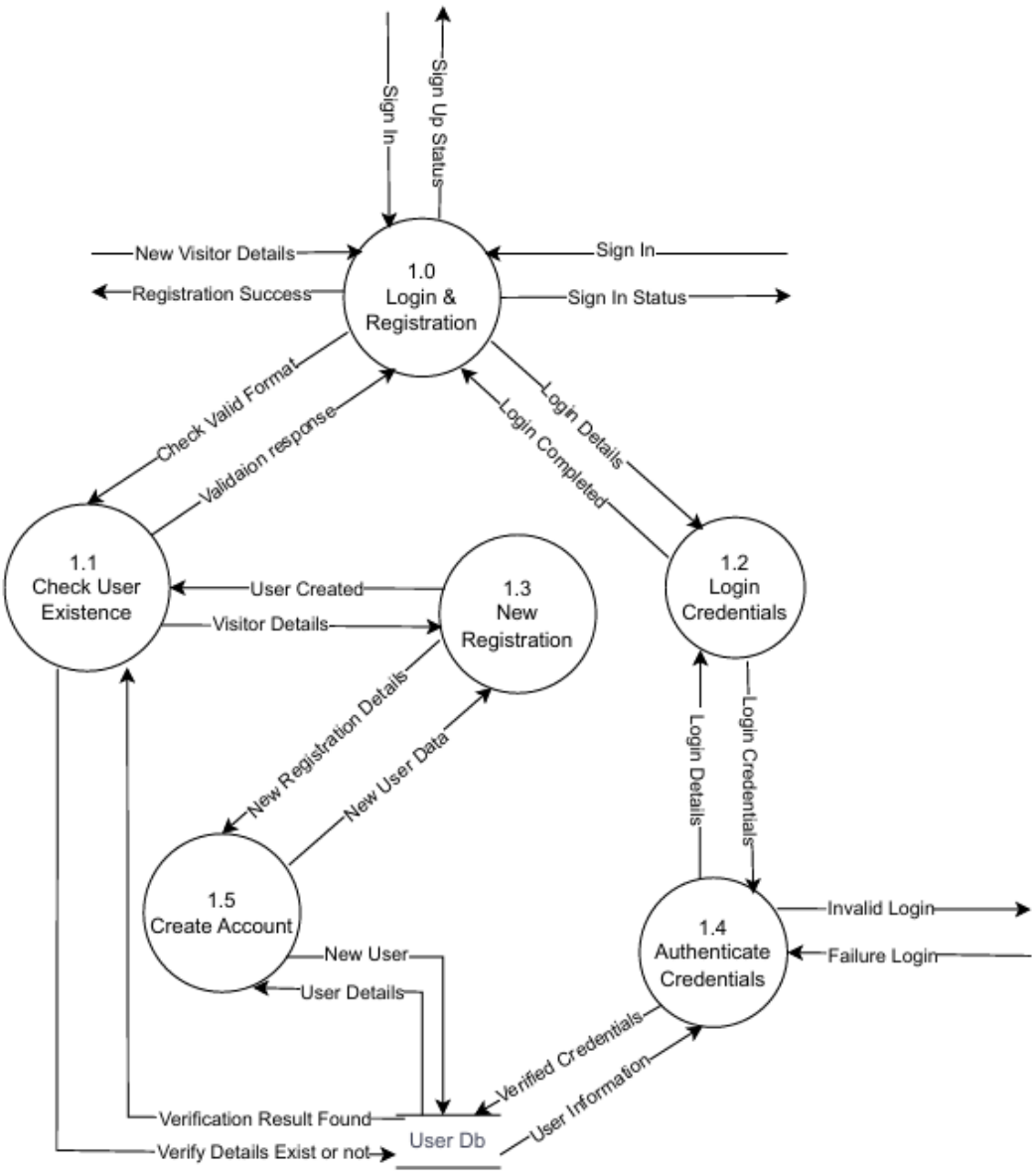


Figure 3 login and registration level 2 DFD

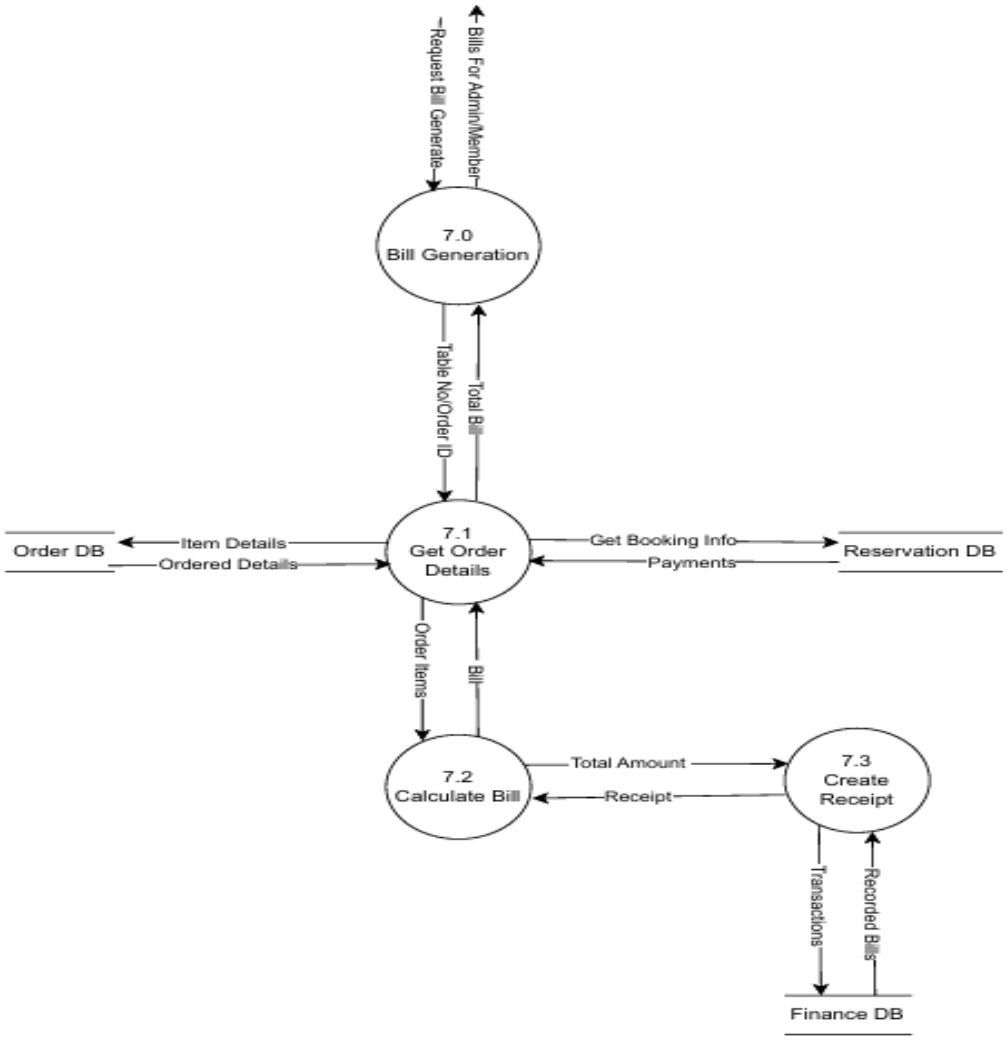


Figure 4 bill generation level 2 DFD

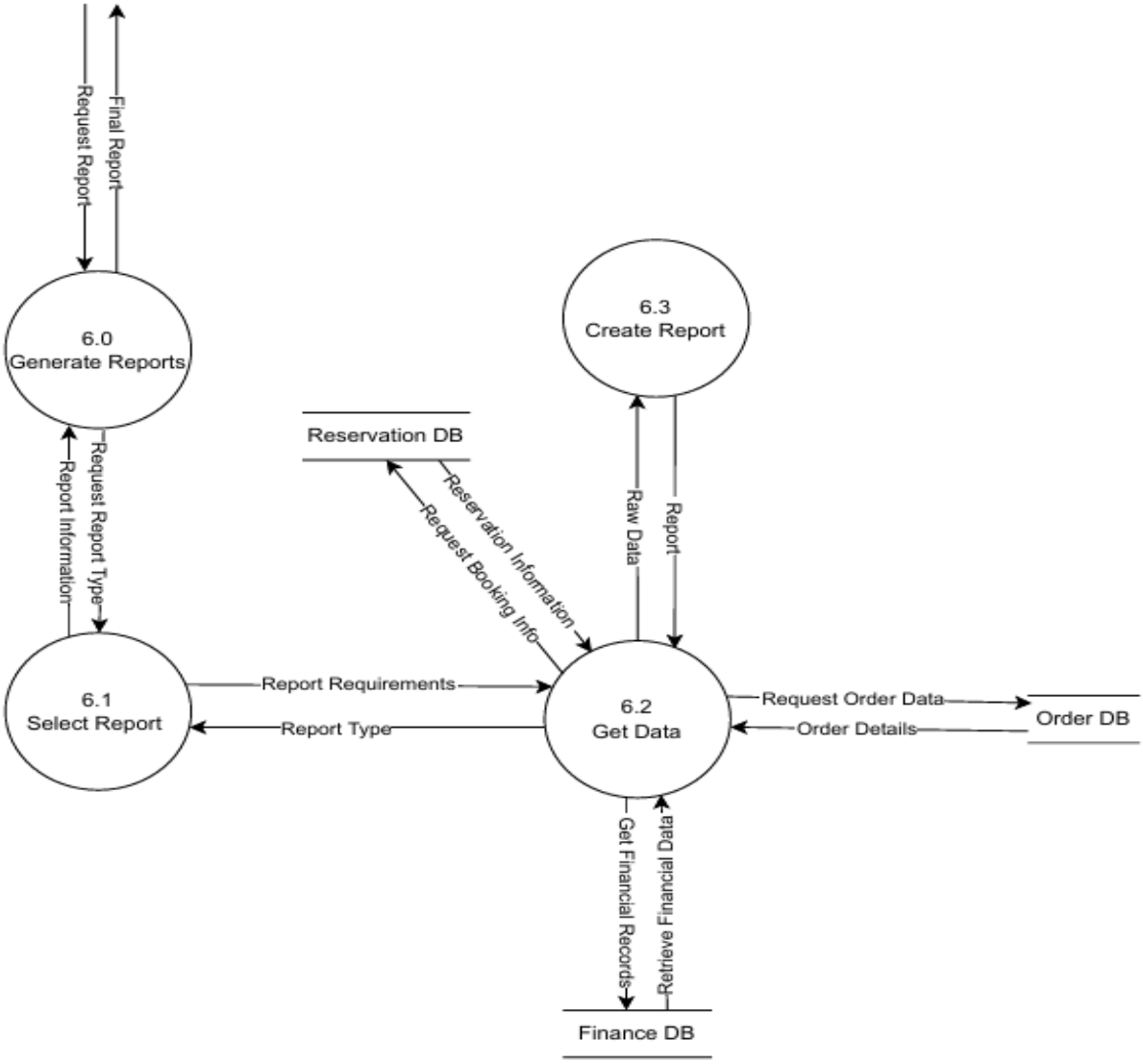


Figure 5 generate reports level 2 DFD

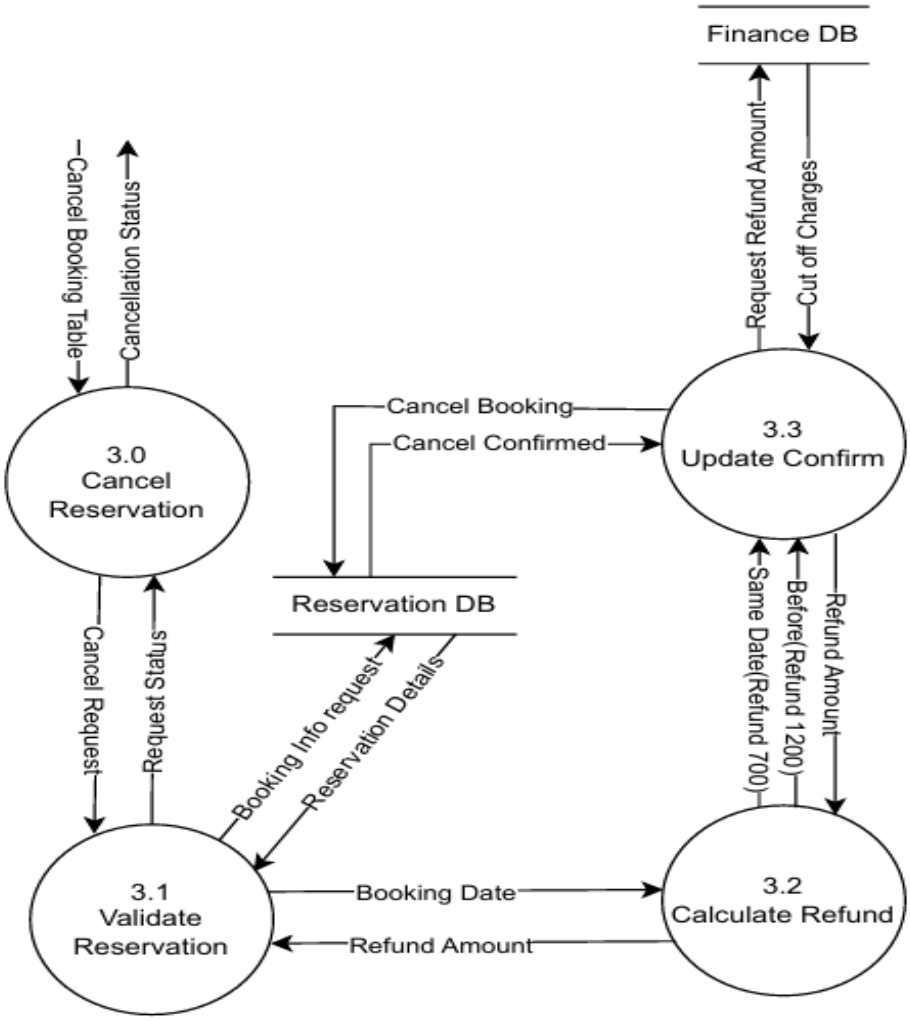


Figure 6 cancel reservations level 2 dfd

Figure 7 post offers and schemes level 2 dfd

2.11.3. Entity Relationship Diagram (ERD)

An Entity-Relationship Diagram (ERD) is a key tool used in database design. It visually shows how a database is structured by representing entities, their attributes and the relationships between them. ERDs help designers understand how data is connected within a system and are commonly used when creating database schemas. They define entities along with their attributes, primary keys and foreign keys. Overall, ER modelling is a technique that represents real-world objects and their relationships in a database (Rodina, 2024). The ERD of our project is presented below:

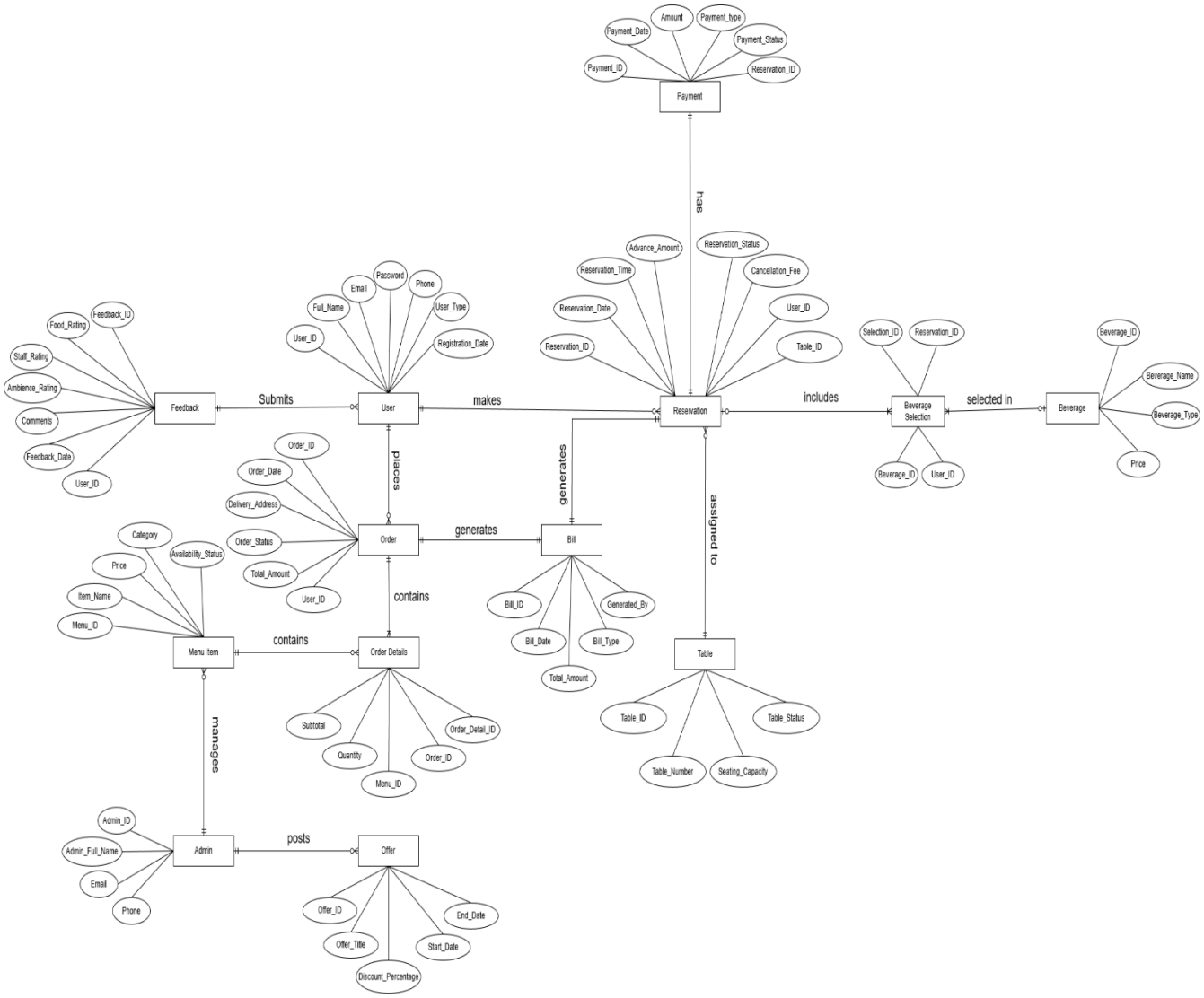


Figure 8 ERD

2.11.4. Data Dictionary

A data dictionary is a document that lists and explains the data elements and field names used in a database. Its main purpose is to provide clear meaning and context for the data, helping database administrators and stakeholders understand the business terms used in the system. It acts as a separate reference document that contains detailed metadata and important information about the database structure and data definitions (Sharma, 2025). The data dictionary of our project is presented below in tabular form.

1. **User_Details =** {User_ID} + Full_Name + [Email] + Password + Phone + User_Type + registration_Date

   User_ID = Integer

   Full_Name = String

   Email = String

   Password = String

   Phone = String

   User_Type = String

   Registration_Date = Date

1. **Table_Details** = {Table_ID} + {Table_Number} + {Seating_Capacity} + {Table _Status}

   Table_ID = Integer

   Table_Number = Integer

   Seating_capacity = Integer

   Table_Status = String

2. **Reservation_Details** = {Reservation_ID} + {Reservation_Date} + {Reservation_Time} + {Advance_Amount} + {Reservation_status} + Cancellation_Fee + User_Id + Table_ID

   Reservation_ID = Integer

   Reservation_Date = Date

   Reservation_Time = Date

   Advance_Amount = Decimal

   Reservation_Status = String

   Cancellation_Fee = Decimal

   User_ID = Integer

   Table_ID = Integer

3. **Payment_Details** = {Payment_ID} + {Payment_Date} + {Amount} + Payment_type +{Payment_Status} + {Reservation_ID}

   Payment_ID = Integer

   Payment_Date = Date

   Amount = Decimal

   Payment_type = String

   Payment_Status = String

   Reservation_ID = Integer

4. **Beverage_Details** = {Beverage_ID} + {Beverage_Name} + {Beverage_Type} + {Price}

   Beverage_ID = Integer

   Beverage_Name = String

   Beverage_Type = String

Price = Decimal

5. **Beverage_Selection** = {Selection_ID} + {Reservation_ID} +{Beverage_ID} + {User_ID}

   Selection_ID = Integer

   Reservation_ID = Integer

   Beverage_ID = Integer

   User_ID = Integer

6. **Menu_Item** ={Menu_ID} + {Item_Name} + {Price} + {Category} + {Availability_Status}

   Menu_ID = Integer

   Item_Name = String

   Price = Decimal

   Category = String

   Availability_Status = String

7. **Order** = {Order_ID} + {Order_Date} + {Delivery_Address} +{Order_status} +{Total_Amount} + User_ID

   Order_ID = Integer

   Order_Date = Date

   Delivery_Address = String

   Order_status = string

   Total_Amount = Decimal

   User_ID = Integer

8. **Order_Details** = {Order_Detail_ID} + {Order_ID} + {Menu_ID} + {Quantity} + {Subtotal}

Order_Detail_ID = Integer

Order_ID = Integer

Menu_ID = Integer

Quantity = Integer

Subtotal = Decimal

9. **Bill_Details** = {Bill_ID} + {Bill_Date} + Total_Amount + {Bill_Type }+ {Generated_By}

   Bill_ID: Integer

   Bill_Date: Date

   Total_Amount: Decimal

   Bill_Type: String

   Generated_By: Integer

10. **Feedback** = {Feedback_ID} + Food_Rating + Staff_Rating + Ambience_Rating + Comments + Feedback_Date + User_ID

    Feedback_ID = Integer

    Food_Rating_ = Integer

    Staff_Rating = Integer

    Ambience_Rating = Integer

    Comments = String

    Feedback_Date = Date

    User_ID = Integer

11. **Offer_Details** = {Offer_ID} + {Offer_Title} + Discount_Percentage + {Start_Date} + {End_Date}

Offer_ID = Integers

Offer_Title = Integer

Discount_Percentage = Decimal

Start_Date = Date

End_Date = Date

12. **Admin** = {Admin_ID} + {Admin_Full_Name} + {Email}, {Phone}

Admin_ID = Integers

Admin_Full_Name = String

Email = String

Phone = String

2.12. Process Specification

A process specification explains the overall steps required to develop a software system. It provides detailed documentation on how the software should be built, including the tasks involved during development, maintenance procedures after completion and guidelines for bug reporting. In this process, testers first identify and test for different types of bugs in the system. They then report these issues to the manager who assigns a team to fix the reported bugs (Geeksforgeeks, 2024).

The process specifications of our project are listed below:

2.12.1. Login and Registration

Table 3 process specification 1.0

Login and Registration	
Process Name	login and registration
Process Number	1.0
Inputs	sign-in request, visitor details, admin login request
Outputs	registration success message, login success or failure message, member access
Process Logic	The process manages user access to the system. It allows new visitors to register and existing users to login. User details are validated, stored and authenticated using the user database. Successful login grants member access, while invalid attempts are rejected.

Table 4 process specification 1.1

Login and Registration	
Process Name	Check user existence
Process Number	1.1
Inputs	visitor details
Outputs	user's existence result
Process Logic	The system checks the user database to determine whether the entered visitor details already exist. If a match is found, the user is treated as an existing user, otherwise the system proceeds with new user registration.

Table 5 process specification 1.2

Login and Registration	
Process Name	Login Credentials
Process Number	1.2
Inputs	username, password
Outputs	login credentials forwarded for authentication
Process Logic	The system accepts login credentials from the user and forwards them to the authentication process for verification against stored user records.

Table 6 process specification 1.3

Login and Registration	
Process Name	New Registration
Process Number	1.3
Inputs	new visitor details
Outputs	new user details
Process Logic	This process collects all required information from a visitor, validates the format and completeness of the data and prepares the information for account creation.

Table 7 process specification 1.4

Login and Registration	
Process Name	Authenticate Credentials
Process Number	1.4
Inputs	login credentials, stored user information
Outputs	login success, invalid login message
Process Logic	The system compares the proceed login credentials with records stored in the user database. If the credentials match, access is granted. If not, a failure message is returned.

Table 8 process specification 1.5

Login and Registration	
Process Name	Create Account
Process Number	1.5
Inputs	new user details
Outputs	user account created, confirmation message
Process Logic	This process creates a new user account using validated visitor details. The information is stored in the user database and the user is notified of successful registration.

2.12.2. Billing generation

Table 9 process specification 7.0

Billing generation	
Process Name	bill generation
Process Number	7.0
Inputs	bill generation request, total bill
Outputs	bill for admin/member, table no/order ID
Process Logic	This process coordinates order retrieval, bill calculation and receipt generation. It ensures all financial data is processed correctly before issuing the final bill.

Table 10 process specification 7.1

Billing generation	
Process Name	Get order details
Process Number	7.1
Inputs	Booking information details
Outputs	ordered items details
Process Logic	The system retrieves order and booking details from the order database and reservation database to prepare for bill calculation.

Table 11 process specification 7.2

Billing generation	
Process Name	Calculate Bill
Process Number	7.2
Inputs	order details
Outputs	total payment amount
Process Logic	This process calculates the total bill amount based on order details, quantities, pricing, taxes and applicable charges.

Table 12 process specification 7.3

Billing generation	
Process Name	Create Receipt
Process Number	7.3
Inputs	Total bill amount
Outputs	Receipt, financial record
Process Logic	A receipt is generated using the final bill amount. Transaction details are stored in the finance database for record-keeping and auditing purposes.

2.12.3. Offers and Schemes Management

Table 13 process specification 10.0

Offers and Schemes Management	
Process Name	Post offers and schemes
Process Number	10.0
Inputs	offer details from admin
Outputs	active offers and schemes
Process Logic	This process manages the creation, storage and publishing of promotional offers and schemes for members.

Table 14 process specification 10.1

Offers and Schemes Management	
Process Name	Create Offer
Process Number	10.1
Inputs	offer scheme details
Outputs	offer information

Process Logic	The admin offers details such as discount type, duration and eligibility. The system prepares this data for storage.

Table 15 process specification 10.2

Offers and Schemes Management	
Process Name	Save Offer
Process Number	10.2
Inputs	offer information
Outputs	stored offer record
Process Logic	The offer details are saved securely in the offers database to ensure persistence and future retrieval.

Table 16 process specification 10.3

Offers and Schemes Management	
Process Name	Publish Offer
Process Number	10.3
Inputs	stored offer data
Outputs	new offer alert, updated interested member list
Process Logic	The system publishes the stored offer and notifies members by sending alerts or updates.

2.12.4. Report Generation Module

Table 17 process specification 6.0

Report Generation Module	
Process Name	Generate Reports
Process Number	6.0
Inputs	request report, request information request
Outputs	final report
Process Logic	This process manages the overall report generation activity. It receives report requests from the user or admin, coordinates report selection, data retrieval and report creation and finally produces the required report.

Table 18 process specification 6.1

Report Generation Module	
Process Name	Select Report
Process Number	6.1
Inputs	Report request
Outputs	report type, report requirements
Process Logic	The user selects the type of report required. The system captures the report requirements and forwards them for data collection.

Table 19 process specification 6.2

Report Generation Module	
Process Name	Get Data
Process Number	6.2
Inputs	Report requirements
Outputs	raw data
Process Logic	This process retrieves required data from multiple databases including reservations DB, order DB and finance DB. The collected data is filtered and prepared for report generation.

Table 20 process specification 6.3

Report Generation Module	
Process Name	Create report
Process Number	6.3
Inputs	raw data
Outputs	generated report
Process Logic	The system processes the collected raw data, formats it according to the selected report type, and generates a structured report for display or download.

2.12.5. Reservation Cancellation Module

Table 21 process specification 3.0

Reservation Cancellation Module	
Process Name	Cancel Reservation
Process Number	3.0
Inputs	cancel request
Outputs	cancellation status
Process Logic	This process handles reservation requests. It verifies booking information, calculates refunds if applicable, updates reservation records and confirms the cancellation to the user.

Table 22 process specification 3.1

Reservation Cancellation Module	
Process Name	Validate Request
Process Number	3.1
Inputs	cancel request, booking information
Outputs	booking date, validate request
Process Logic	The system validates the cancellation request by checking booking details and cancellation rules. Only valid requests are forwarded for refund calculation.

Table 23 process specification 3.2

Reservation Cancellation Module	
Process Name	Calculate Refund
Process Number	3.2
Inputs	booking date
Outputs	refund amount
Process Logic	This process calculates the refundable amount based on booking date, cancellation policy and applicable charges or deductions.

Table 24 process specification 3.3

Reservation Cancellation Module	
Process Name	Update Confirm
Process Number	3.3
Inputs	refund amount, cancel confirmation
Outputs	updated reservation record, refund request
Process Logic	The system updates the reservation status in the reservation database, records refund details in the finance database and confirms the cancellation and refund to the user.

2.13. Design specification

2.13.1. Structure chart

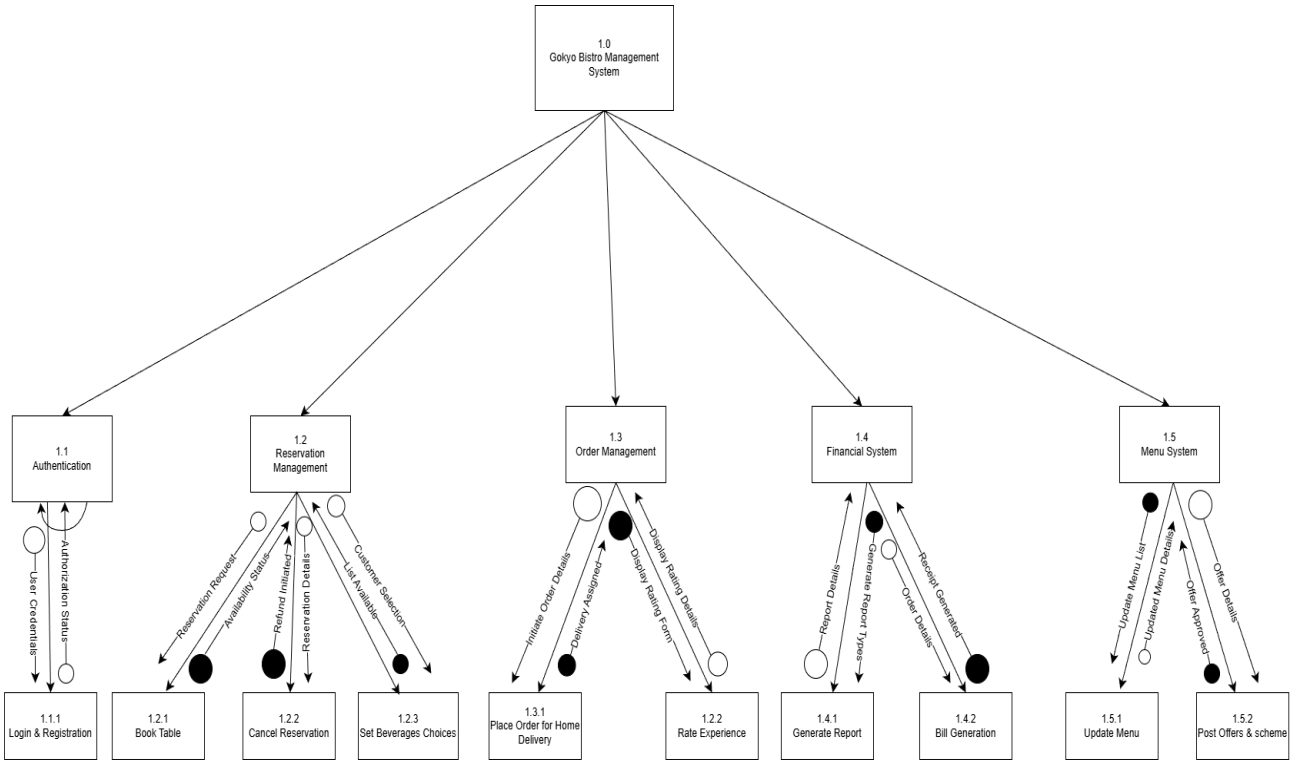


Figure 9 structure chart

2.14. Progress logs

Progress Log

Meeting No: 01 **Date: 12/12/2025**

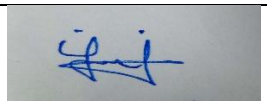
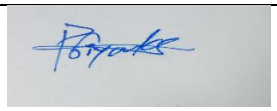
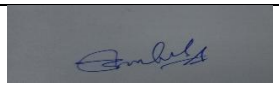
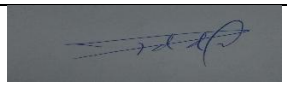
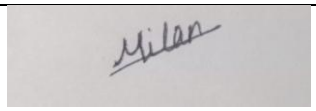
Start Time: 10:50 am **Finish Time: 11:00 am**

Items Discussed: Work Distribution
Introduction: Pankaj Rai
DFD's (level 0, level 1, and level 2): Ejikeyal Pakhrin
Data Dictionary: Palden Lama
SRS: Priyanka Kumari Shah
Business Case and Process Specification: Milan Lama and Pankaj Rai
ER diagram: Ejikeyal Pakhrin and Palden Lama

Achievement: Nothing achieved.

Problems (if any): No problem faced yet.

Tasks for Next Meeting: Start research for the task divided.

Student's Name	Student's Signature
Pankaj Rai (C11)	
Palden Lama (C11)	
Ejikeyal Pakhrin (C11)	
Priyanka Kumari Shah (C11)	
Milan Lama (C11)	

Progress Log

Meeting No: 02 **Date: 19/12/2025**

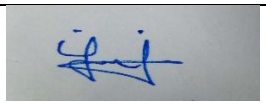
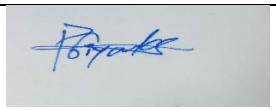
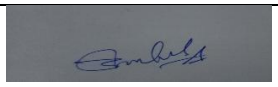
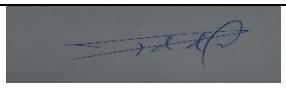
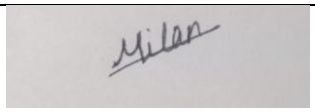
Start Time: 10:45 am **Finish Time: 11:20 am**

Items Discussed: Work Progress
Work Progress details of each task. And Check if any problem has occurred. Then investigate for the problems that have occurred.

Achievement: Completed process specification.
Finished Introduction.
Finished Data Dictionary.

Problems (if any): Faced some issue in DFD. Trying to ensure that DFDs were balanced of not.

Tasks for Next Meeting: Fix the issues that have occurred and complete the remaining task for the upcoming milestone submission.

Student's Name	Student's Signature
Pankaj Rai (C11)	
Palden Lama (C11)	
Ejikeyal Pakhrin (C11)	
Priyanka Kumari Shah (C11)	
Milan Lama (C11)	

Progress Log

Meeting No: 03 **Date: 22/12/2025**

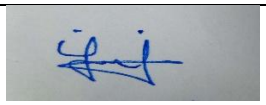
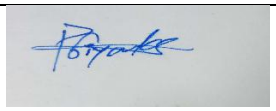
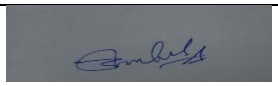
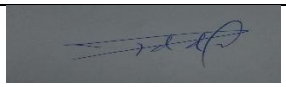
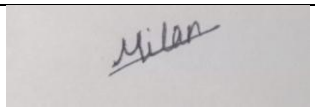
Start Time: 2:00 pm **Finish Time: 4:00 am**

Items Discussed: Task completed or not. And previously faced problem resolved or not. And prepare for submission.

Achievement: Resolved the DFD problem, and properly developed DFD context diagram and decomposed it to level 1 and level 2.
Created ERD.
Finished Business Case and SRS.
Milestone Submitted

Problems (if any): Challenges if business case and ERD were correct or not.

Tasks for Next Meeting: Prepare for the task division for the next upcoming milestone 2.

Student's Name	Student's Signature
Pankaj Rai (C11)	
Palden Lama (C11)	
Ejikeyal Pakhrin (C11)	
Priyanka Kumari Shah (C11)	
Milan Lama (C11)	

Progress Log

Meeting No: 04 **Date: 28/12/2025**

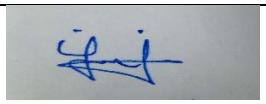
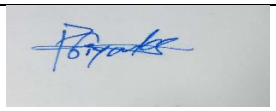
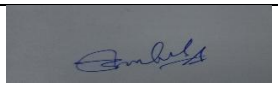
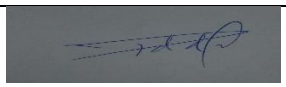
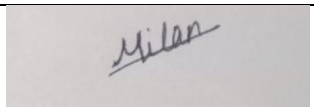
Start Time: 9:10 am **Finish Time: 9:20 am**

Items Discussed: Review of Milestone 1. And dividing individual task for milestone 2.
Book Table and Group DFD: Priyanka Kumari Shah
Rate the experience and Group Structure Chart: Ejikeyal Pakhrin
Set Beverage Choices and Conclusion: Pankaj Rai
Update Menu and Progress Log: Milan Lama
Place Order for Home Delivery and Group DFDs: Palden Lama

Achievement: Individual Task assigned to each member.
Updating the progress log.

Problems (if any): No problem faced yet.

Tasks for Next Meeting: Each member should do their individual task. Resolve issues found in milestone 1.

Student's Name	Student's Signature
Pankaj Rai (C11)	
Palden Lama (C11)	
Ejikeyal Pakhrin (C11)	
Priyanka Kumari Shah (C11)	
Milan Lama (C11)	

Progress Log

Meeting No: 05 **Date: 04/01/2026**

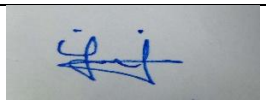
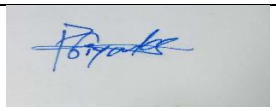
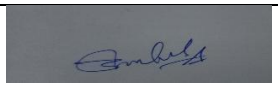
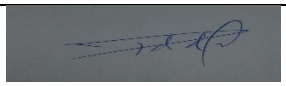
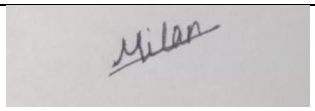
Start Time: 9:00 am **Finish Time: 9:50 am**

Items Discussed: Work Progress and Refinement
Work Progress details of each individual tasks. And Check if any problem has occurred. Then investigate for the problems that have occurred. Previous milestone 1 problem fixed or not.

Achievement: Draft for specific Module Specifications and detailed DFDS for chosen module.
Updating Progress Log.
Writing conclusion and finalizing format for submission.

Problems (if any): Resolving issue during dfd level 2 and structure chart.

Tasks for Next Meeting: Properly format and finalize individuals' task for submission.

Student's Name	Student's Signature
Pankaj Rai (C11)	
Palden Lama (C11)	
Ejikeyal Pakhrin (C11)	
Priyanka Kumari Shah (C11)	
Milan Lama (C11)	

3. Individual task

3.1. 2456 Pankaj Rai

Student Name: Pankaj Rai

LMU ID: 2456

Function chosen: Set Beverage Choices

Function Number: 4

3.1.1. Context Level

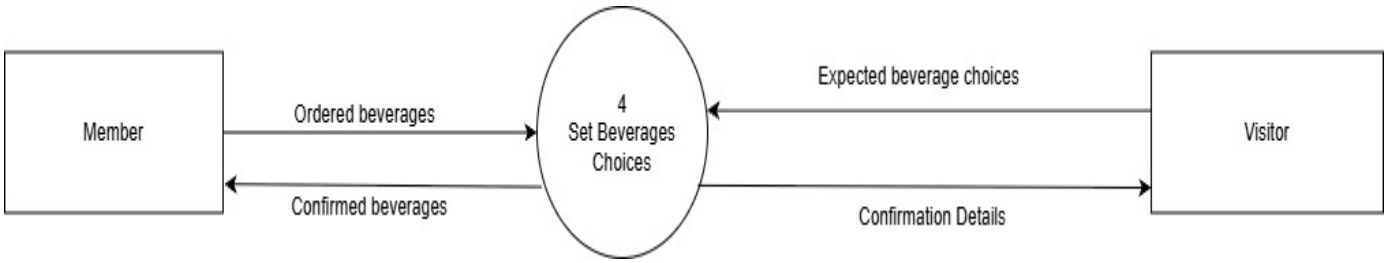


Figure 10 set beverages choices context diagram

3.1.2. Level 1 DFD fragments

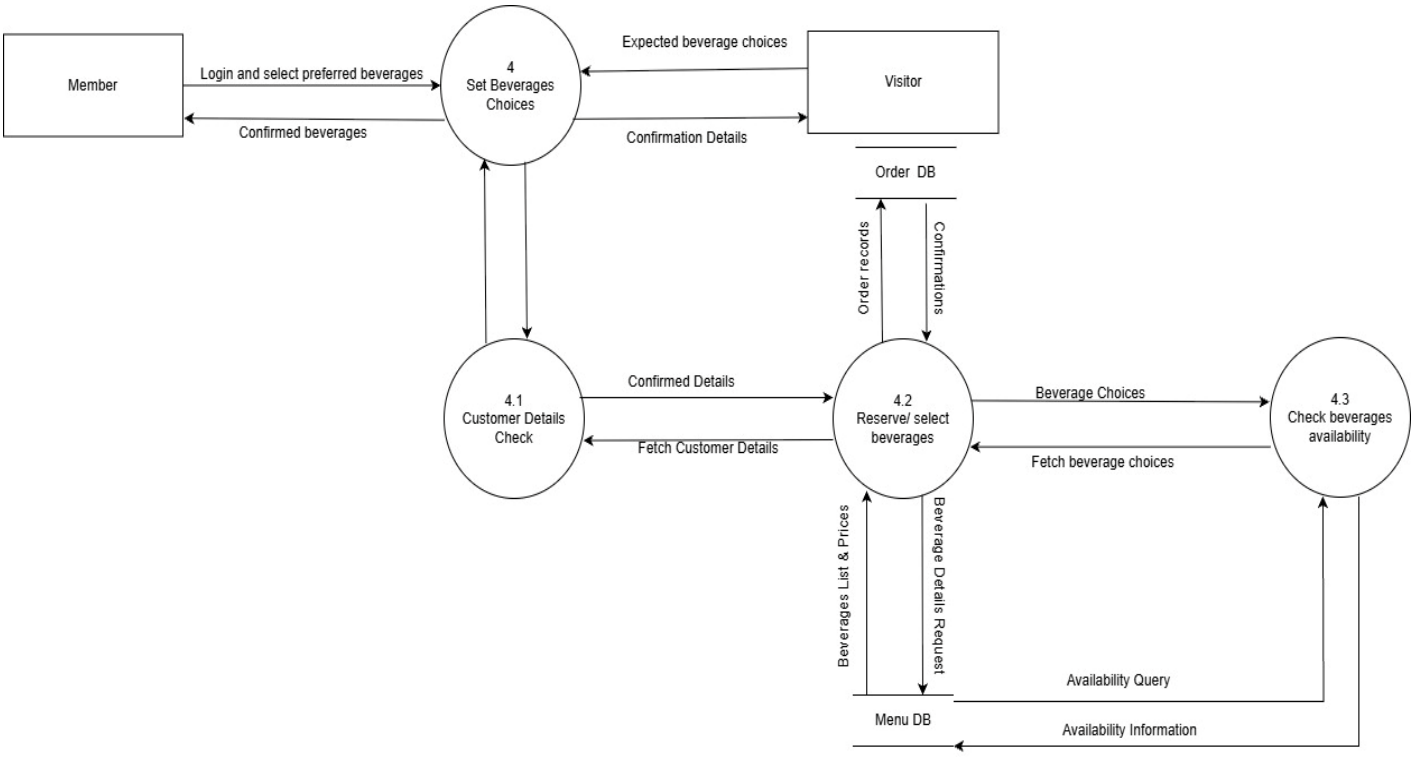


Figure 11 set beverages choices DFD 1

3.1.3. Level 2 DFDs for the particular function

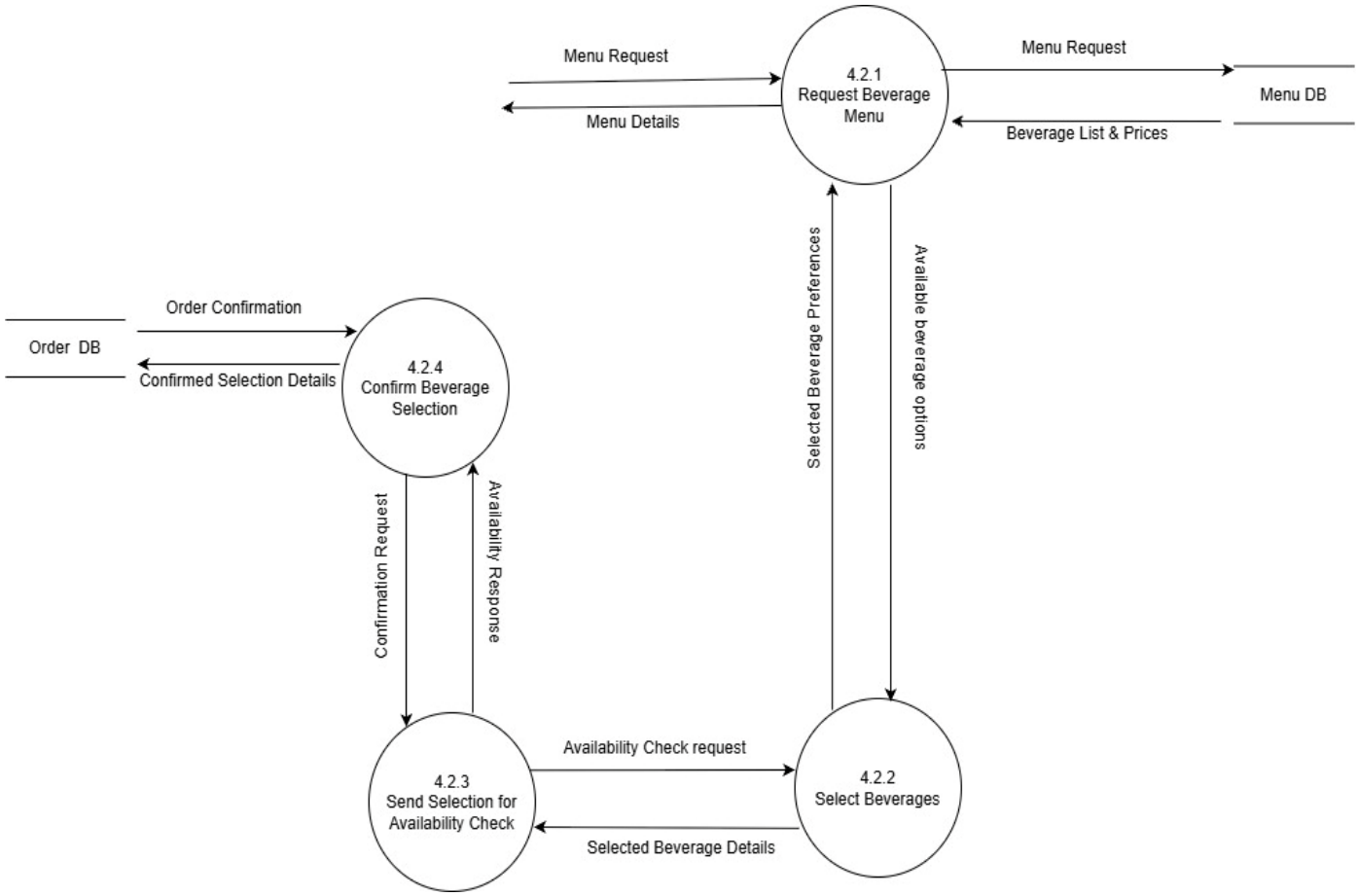


Figure 12 set beverages choices DFD

3.1.4. Structure chart

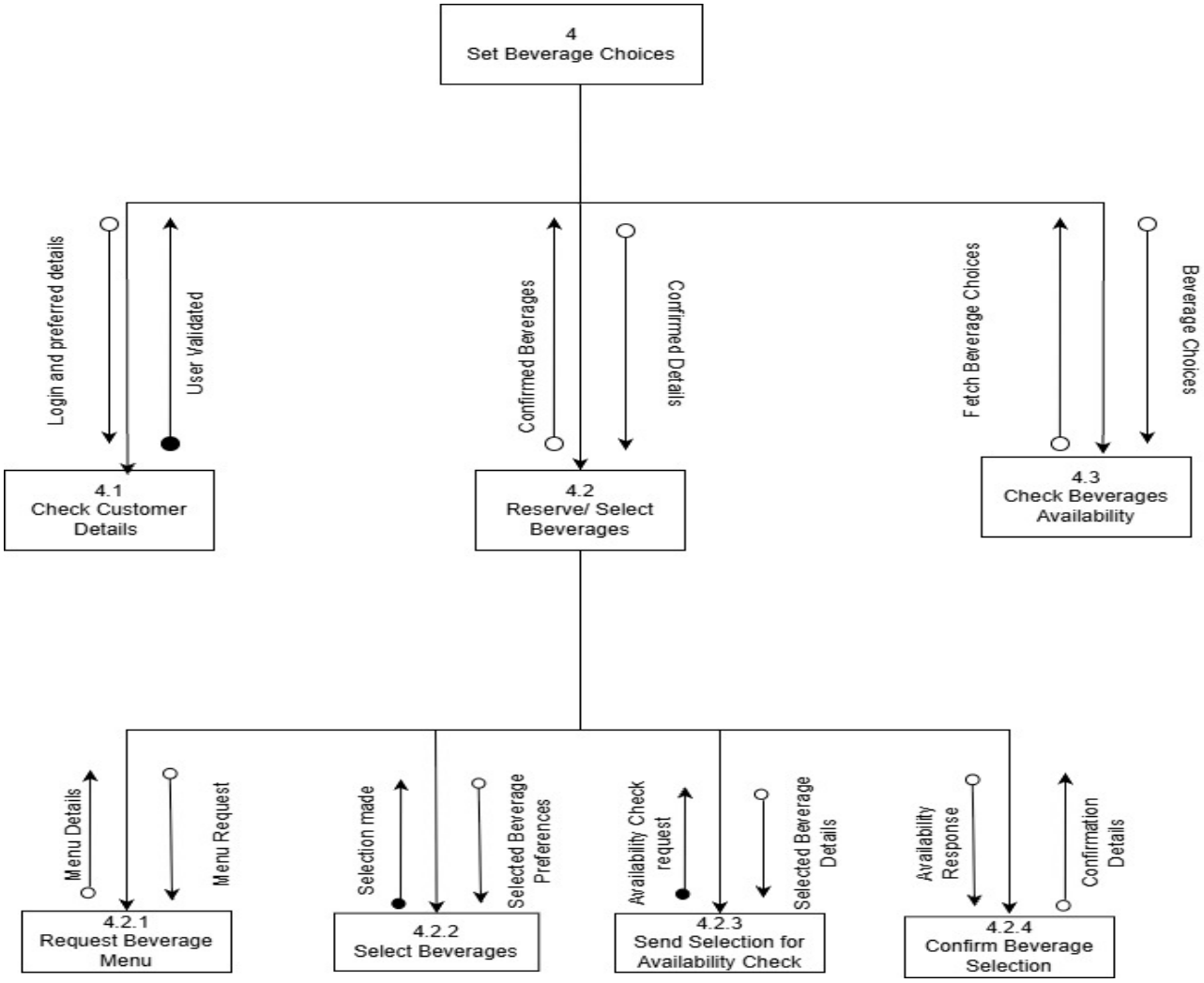


Figure 13 set beverages structure chart

3.1.5. Module specifications

MODULE NAME: Select beverages

MODULE NUMBER: 4.2.2

PURPOSE: To coordinate the retrieval of the beverage menu, capture user preferences, and finalize the selection after availability is verified.

PSEUDOCODE:

```
CALL Request Beverage Menu.
RECEIVE Menu Details from Menu DB.
CALL Select Beverages to capture Selected Beverage Preferences.
IF Selection is made THEN
   CALL Send Selection for Availability Check.
   RECEIVE Availability response.
   IF Availability response is "Available" THEN
      CALL Confirm Beverage Selection.
      SET Confirmed Details = Selection summary.
   ELSE
      SET Error_Msg = "Beverage out of stock".
   ENDIF
ENDIF
RETURN Confirmed Details and Confirmation Details.
```

INPUT PARAMETERS:

- Menu Request
- Selected Beverage Preferences
- Availability response

OUTPUT PARAMETERS:

- Confirmed Details
- Confirmation Details

GLOBAL VARIABLES: None

LOCAL VARIABLES:

- Selection_Summary
- Availability_Status

CALLS:

- Request Beverage Menu
- Select Beverages
- Send Selection for Availability Check
- Confirm Beverage Selection

CALLED BY: Get Beverage Choices

3.2. 2449 Eijkeyal Pakhrin

Student Name: Eijkeyal Pakhrin

LMU ID: 2449

Function chosen: Rate the Experience

Function Number: 9

3.2.1. Context Level

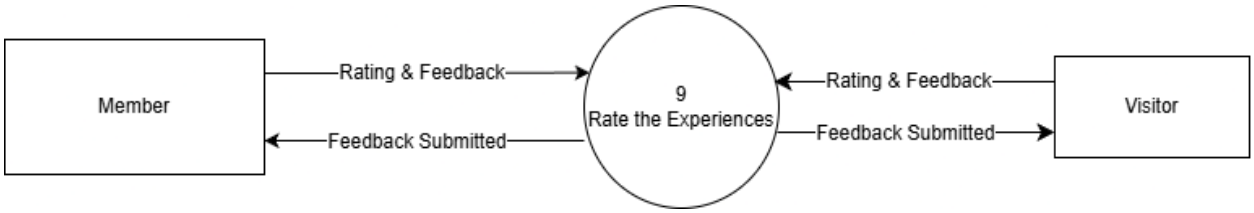


Figure 14 rate the experience context diagram

3.2.2. Level 1 DFD fragments

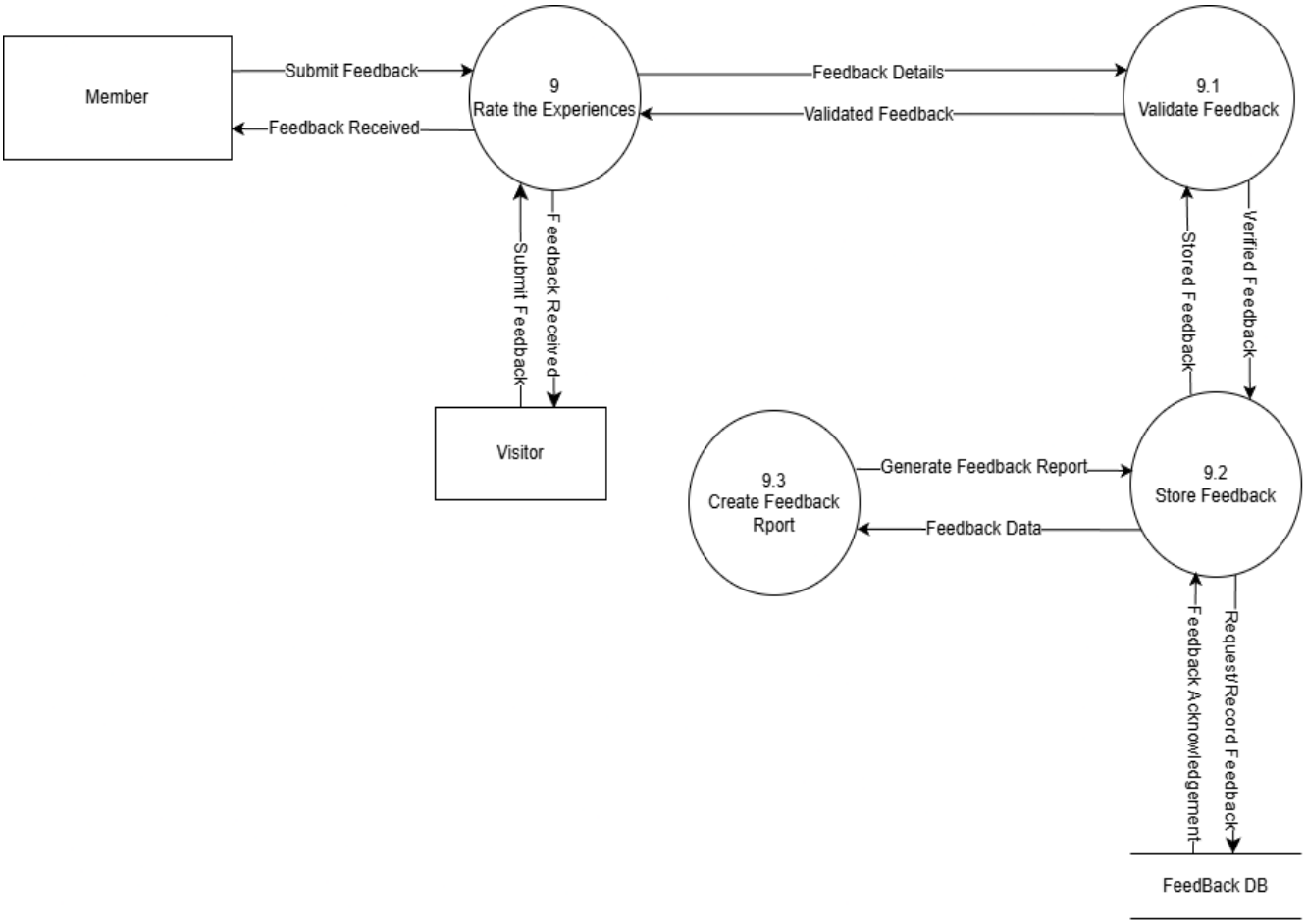


Figure 15 rate the experience DFD 1

3.2.3. Level 2 DFDs for the particular function

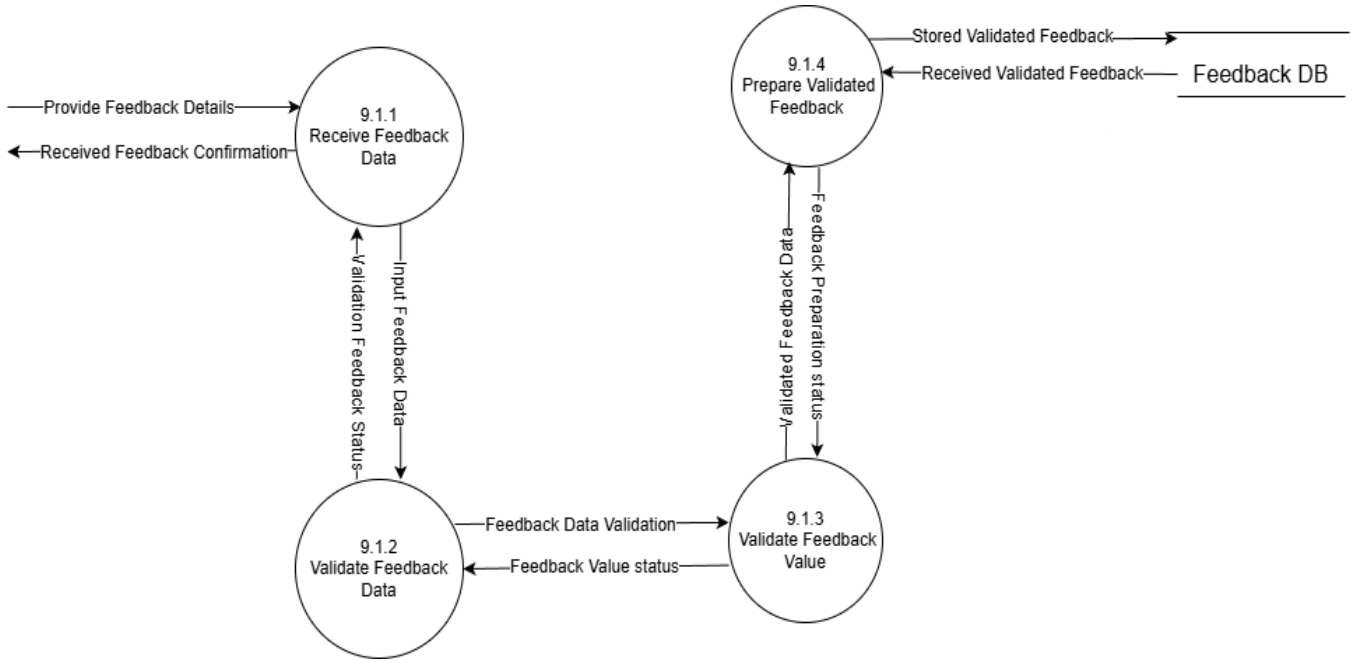


Figure 16 rate the experience DFD 2

3.2.4. Structure chart

A structure chart is a diagram used to show the logical flow of processes in a system. It follows an inverted tree format where each box represents a module or function. The main system task is broken down into smaller submodules, making the design easier to understand and manage. This modular approach helps teams divide work efficiently and allows each module to be tested independently. It also gives a clear hierarchical view of how different functions operate within the system.

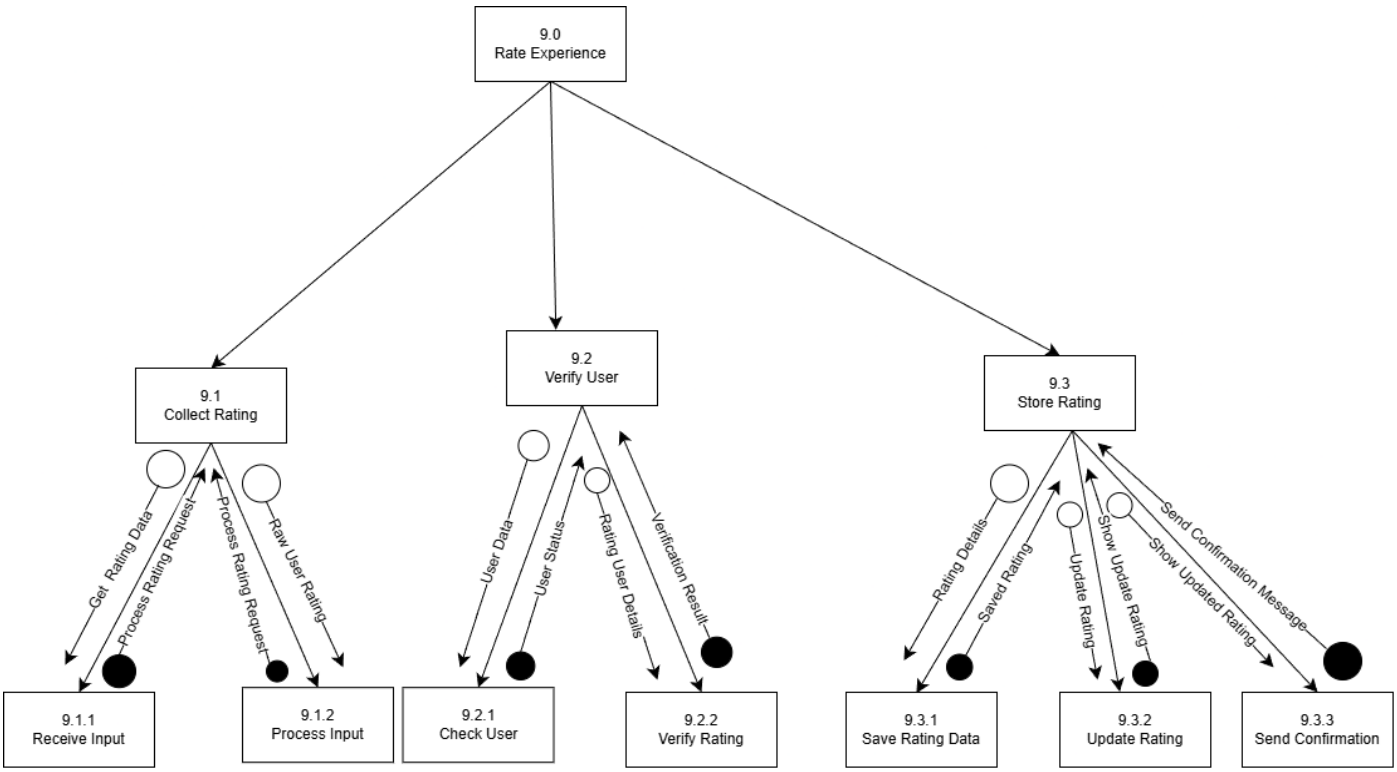


Figure 17 rate the experience structure chart

3.2.5. Module specifications

MODULE NAME: Verify Rating

MODULE NUMBER: 9.2.2

PURPOSE: To ensure that the feedback and rating provided by the Member is complete, accurate, and follows business rules before it is passed to the storage module.

PSEUDOCODE:

```
CALL Receive Rating Data to get feedback details.
IF Rating Data is received THEN
   CALL Validate Rating Data to check for completeness.
   IF Completeness Status is True THEN
      CALL Validate Rating Value with complete rating data.
      IF Validation Status is Valid THEN
         CALL Prepare Validated Rating.
         SET Storage_Detail = Validated Rating.
      ELSE
         SET Error_Msg = "Invalid rating value".
      ENDIF
   ELSE
      SET Error_Msg = "Incomplete rating data".
   ENDIF
ENDIF
RETURN Storage_detail and Validated Rating to Store Rating.
```

INPUT PARAMETERS:

- Feedback details
- Rating Data
- Validation Status

OUTPUT PARAMETERS:

- Rating confirmation
- Storage detail
- Validated Rating

GLOBAL VARIABLES: None

LOCAL VARIABLES:

- Completeness_Status
- Preparation_Status

CALLS:

- Receive Rating Data
- Validate Rating Data
- Validate Rating Value
- Prepare Validated Rating

CALLED BY: Rate the Experiences

3.3. 2454 Palden Lama

Student Name: Palden Lama

LMU ID: 2454

Function chosen: Place Order for Home Delivery

Function Number: 5

3.3.1. Context Level

Figure 18 place order for home delivery context diagram

3.3.2. Level 1 DFD fragments

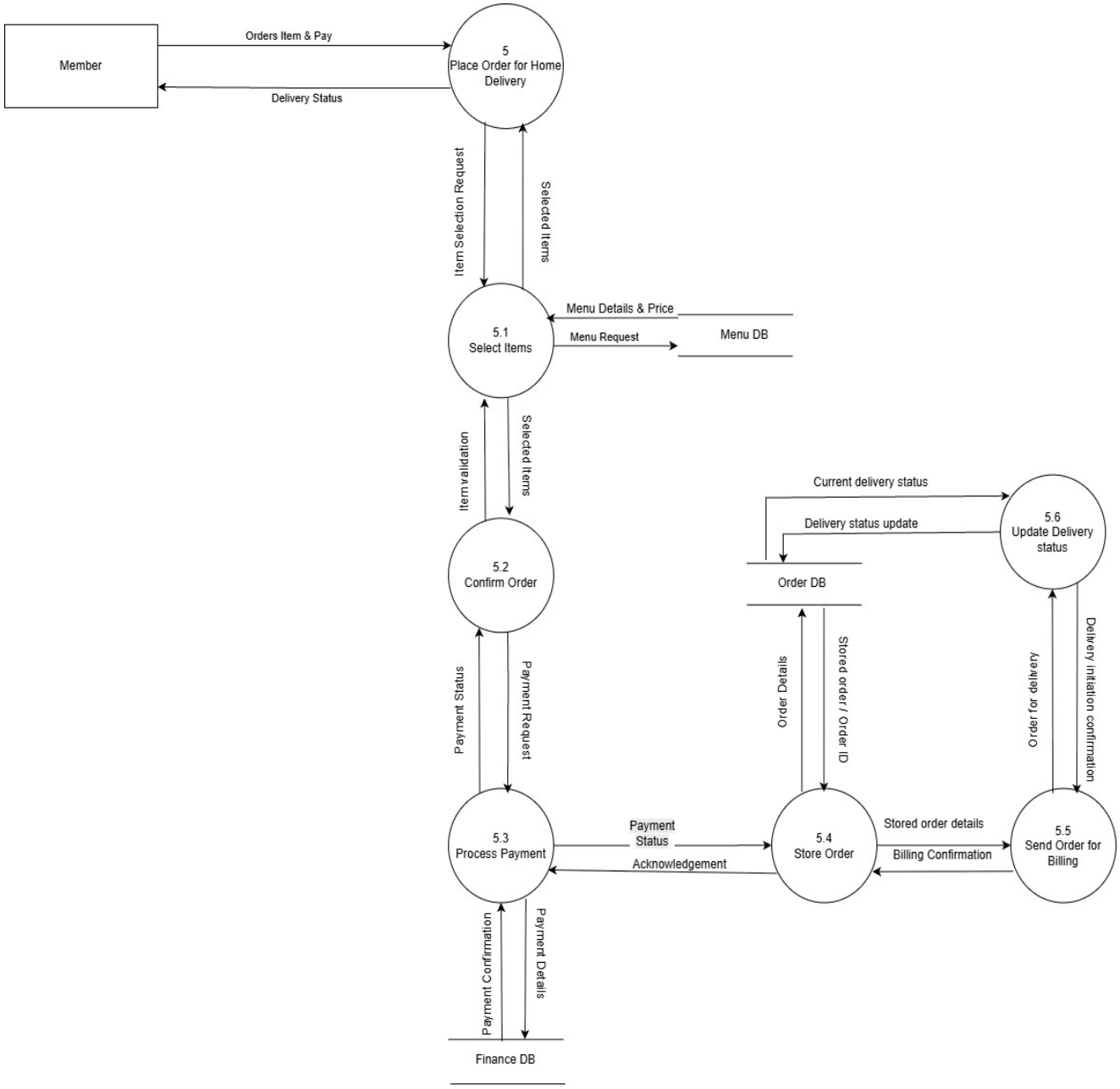


Figure 19 place order for home delivery DFD 1

3.3.3. Level 2 DFDs for the particular function

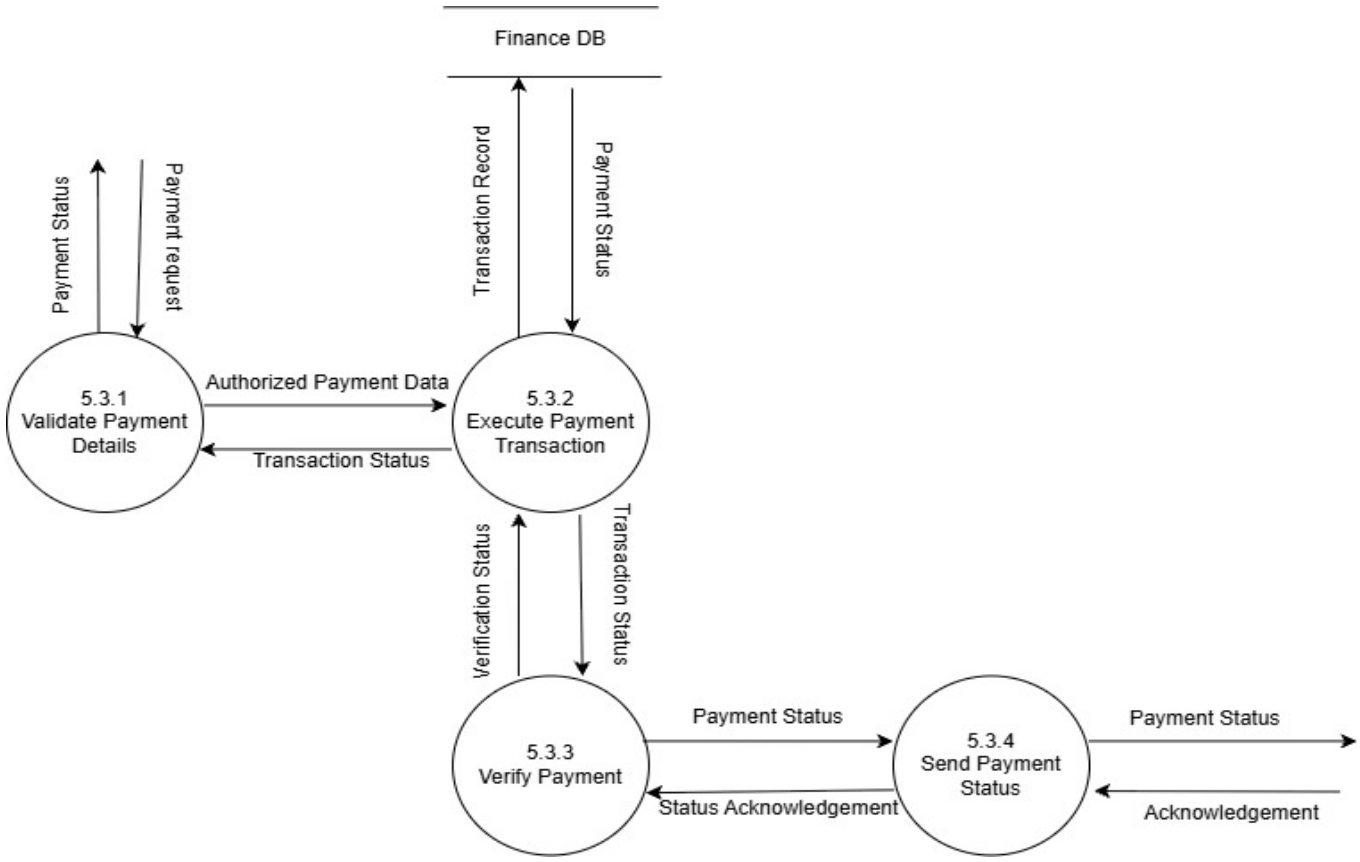


Figure 20 place order for home delivery DFD 2

3.3.4. Structure chart

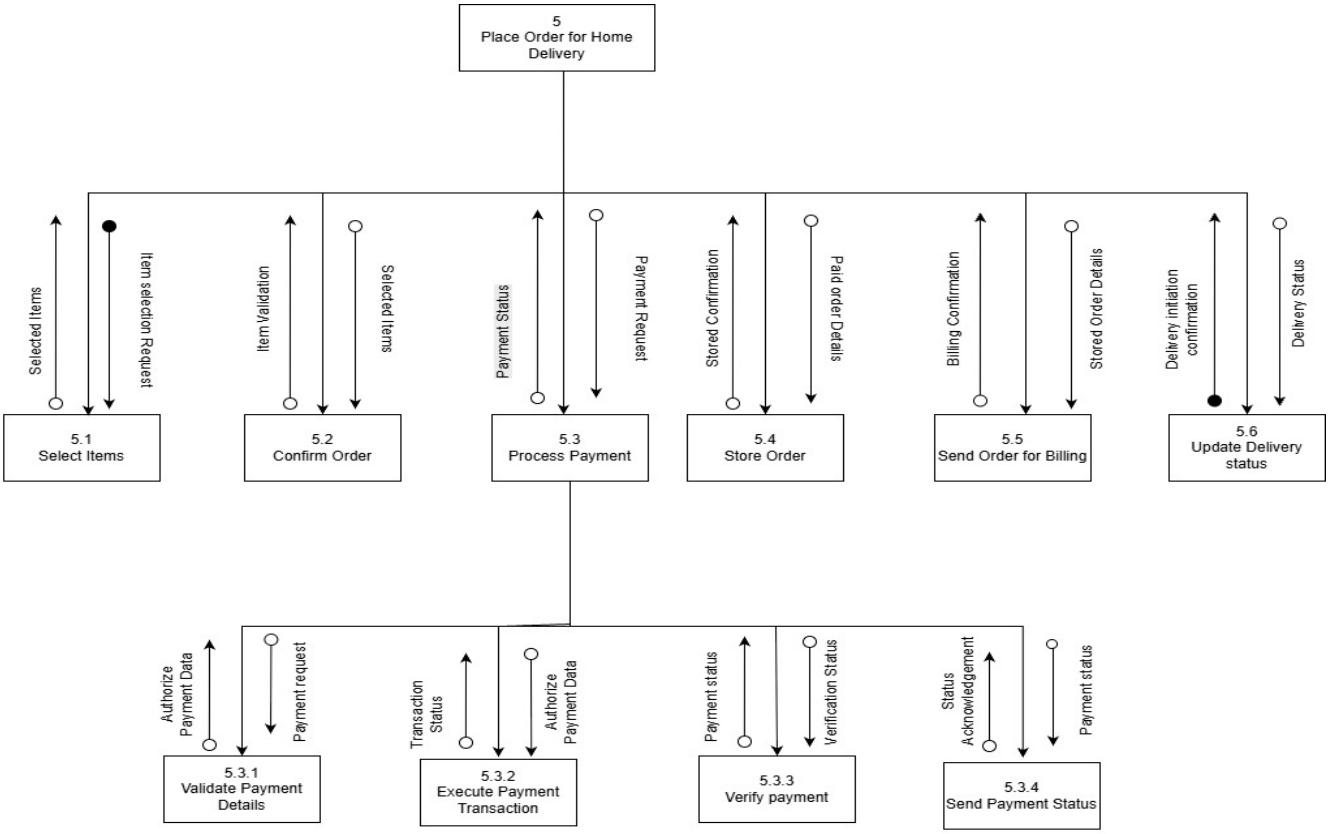


Figure 21 place order for home delivery structure chart

3.3.5. Module specifications

MODULE NAME: Process Payment

MODULE NUMBER: 5.3

PURPOSE: To coordinate the validation, execution, and verification of a member's payment before finalizing the home delivery order.

PSEUDOCODE:

```
RECEIVE Payment Details from Member.
CALL Validate Payment Details with Payment Request.
IF Payment_Data is Authorized THEN
   CALL Execute Payment Transaction to interface with Finance DB.
   RECEIVE Transaction Status.
   CALL Verify Payment with Verification Status.
   IF Payment is Successful THEN
      SET Payment_Status = "Success".
      SEND Paid Order Details to Store Order.
   ELSE
      SET Payment_Status = "Failed".
   ENDIF
ENDIF
CALL Send Payment Status to return status to Member.
RETURN Payment Confirmation.
```

INPUT PARAMETERS:

- Payment Details
- Payment Request

OUTPUT PARAMETERS:

- Payment Status
- Payment Confirmation
- Paid Order Details

GLOBAL VARIABLES: None

LOCAL VARIABLES:

- Authorization_Flag

- Transaction_Status

CALLS:

- Validate Payment Details
- Execute Payment Transaction
- Verify Payment
- Send Payment Status

CALLED BY:

- Place Order for Home Delivery

3.4. 2458 Priyanka Kumari Shah

Student Name: Priyanka Kumari Shah

LMU ID: 2458

Function chosen: Book Table

Function Number: 2

3.4.1. Context Level

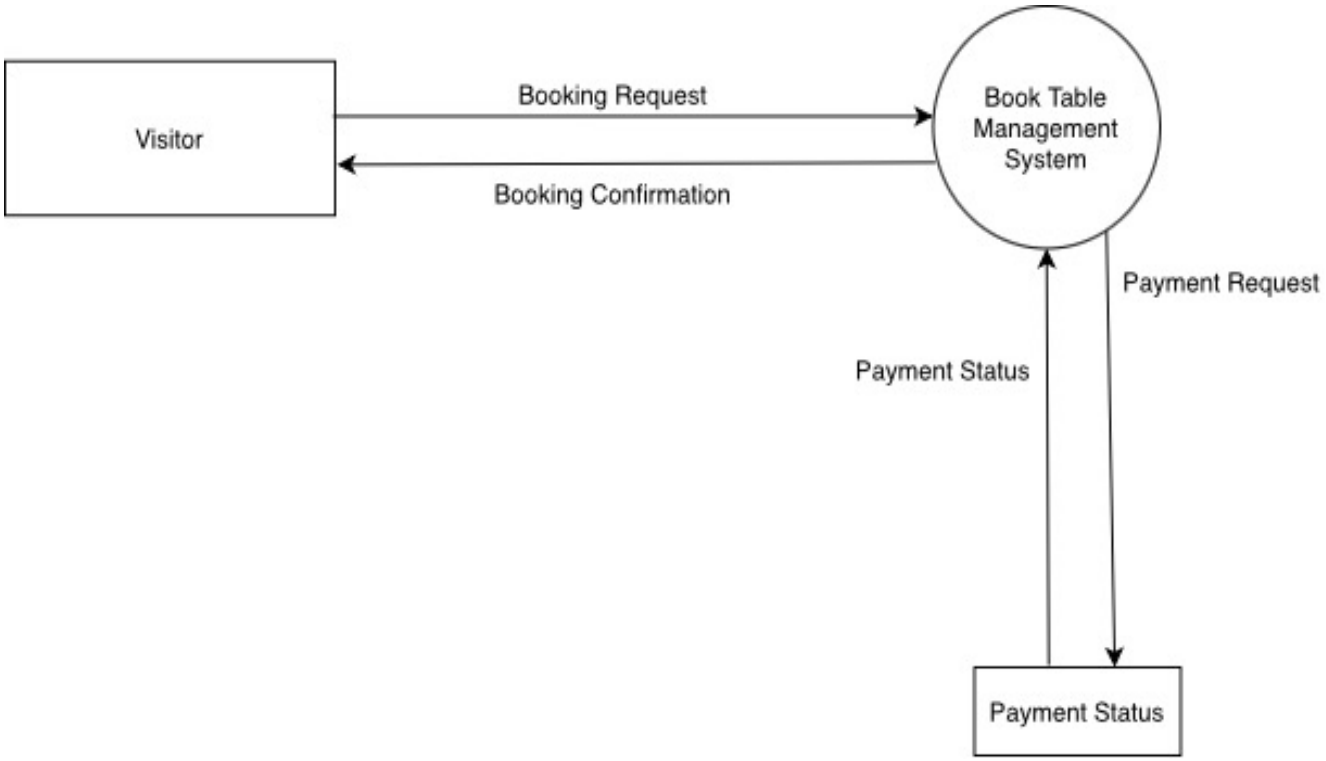


Figure 22 book table context diagram

3.4.2. Level 1 DFD fragments

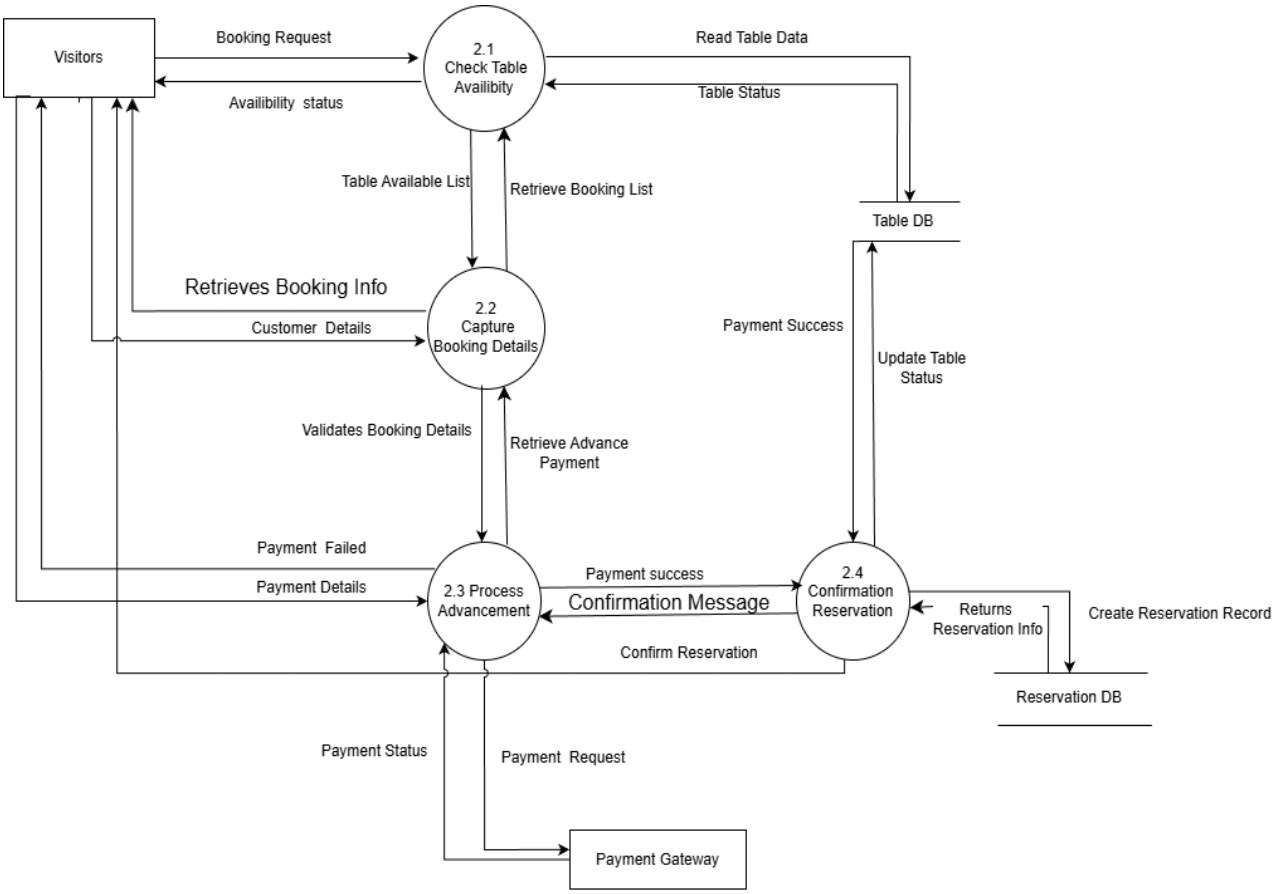


Figure 23 book table DFD 1

3.4.3. Level 2 DFDs for the particular function

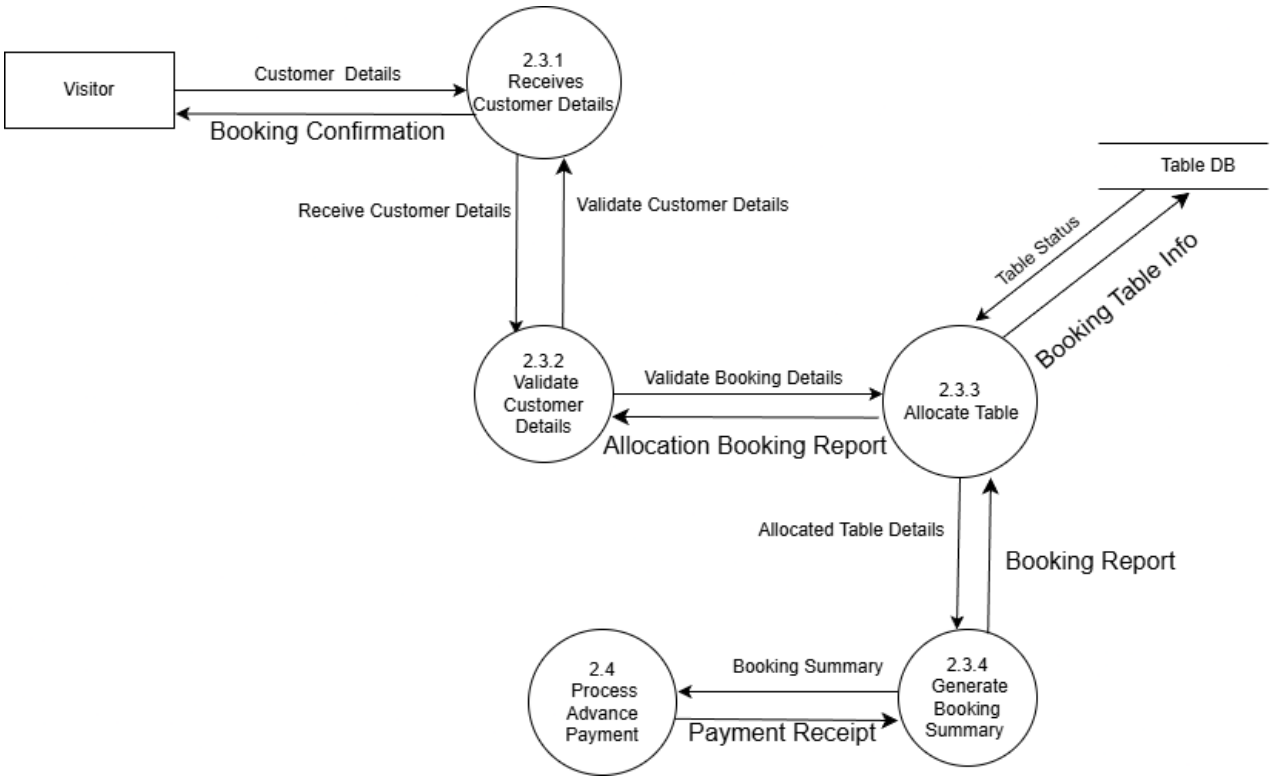


Figure 24 book table DFD 2

3.4.4. Structure chart

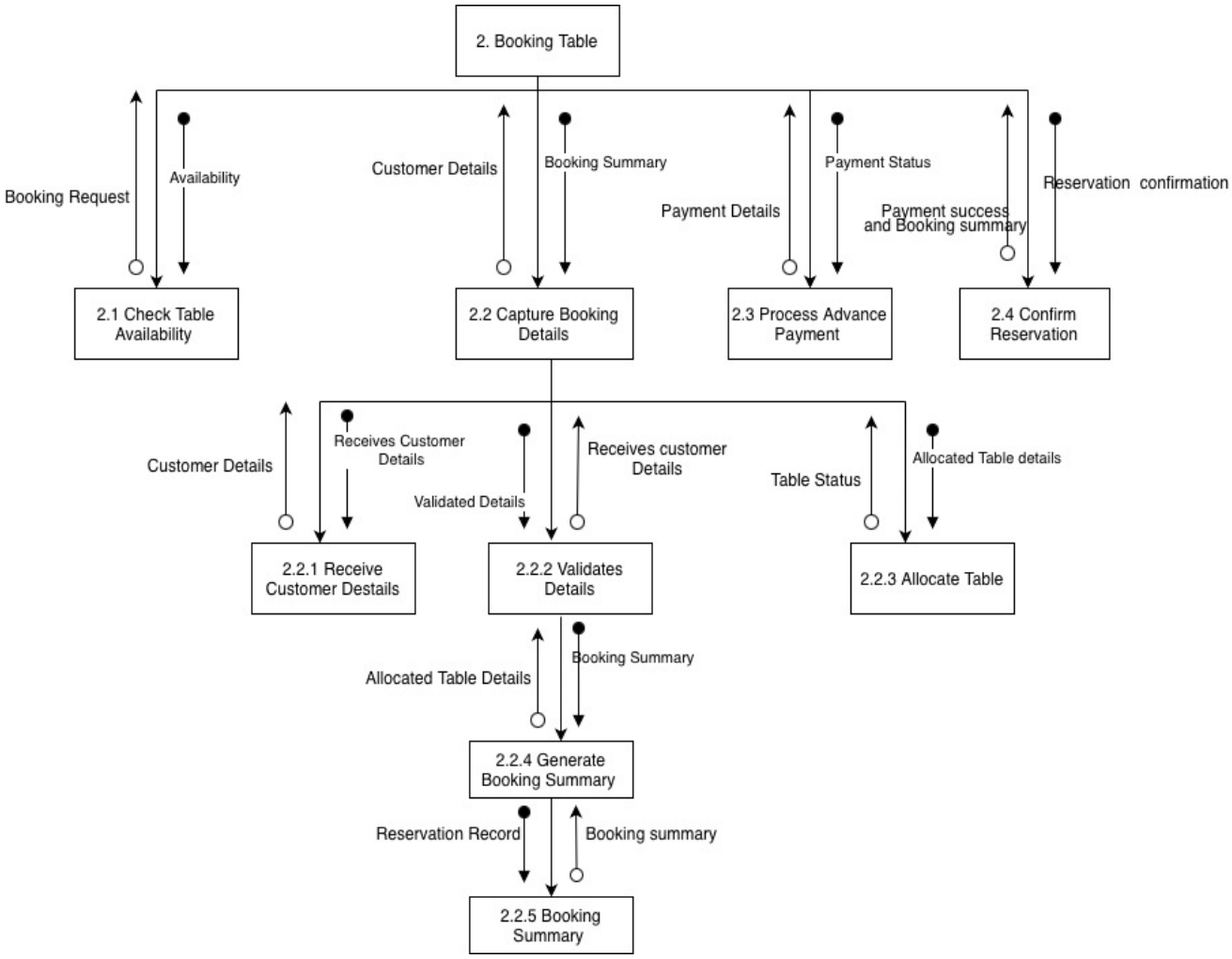


Figure 25 book table structure chart

3.4.5. Module specifications

Module Name: Check Table Availability

Purpose : Check whether the table are available for the selected date and time.

Module Number: 2.1

Pseudocode: Receive date and time

Check table database

If table available

Return available

Else

Return not available

Input Parameters: date and time

Output Parameters: available tables

Global Variables: table database

Local Variables: available tables

Calls: none

Called By: BookTableController

3.5. 2452 Milan Lama

Student Name: Milan Lama

LMU ID: 2452

Function chosen: Update Menu

Function Number: 8

3.5.1. Context Level

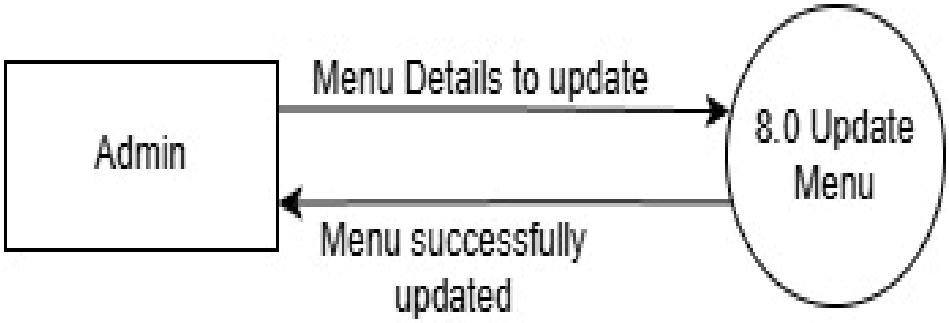


Figure 26 update menu context level

3.5.2. Level 1 DFD fragments

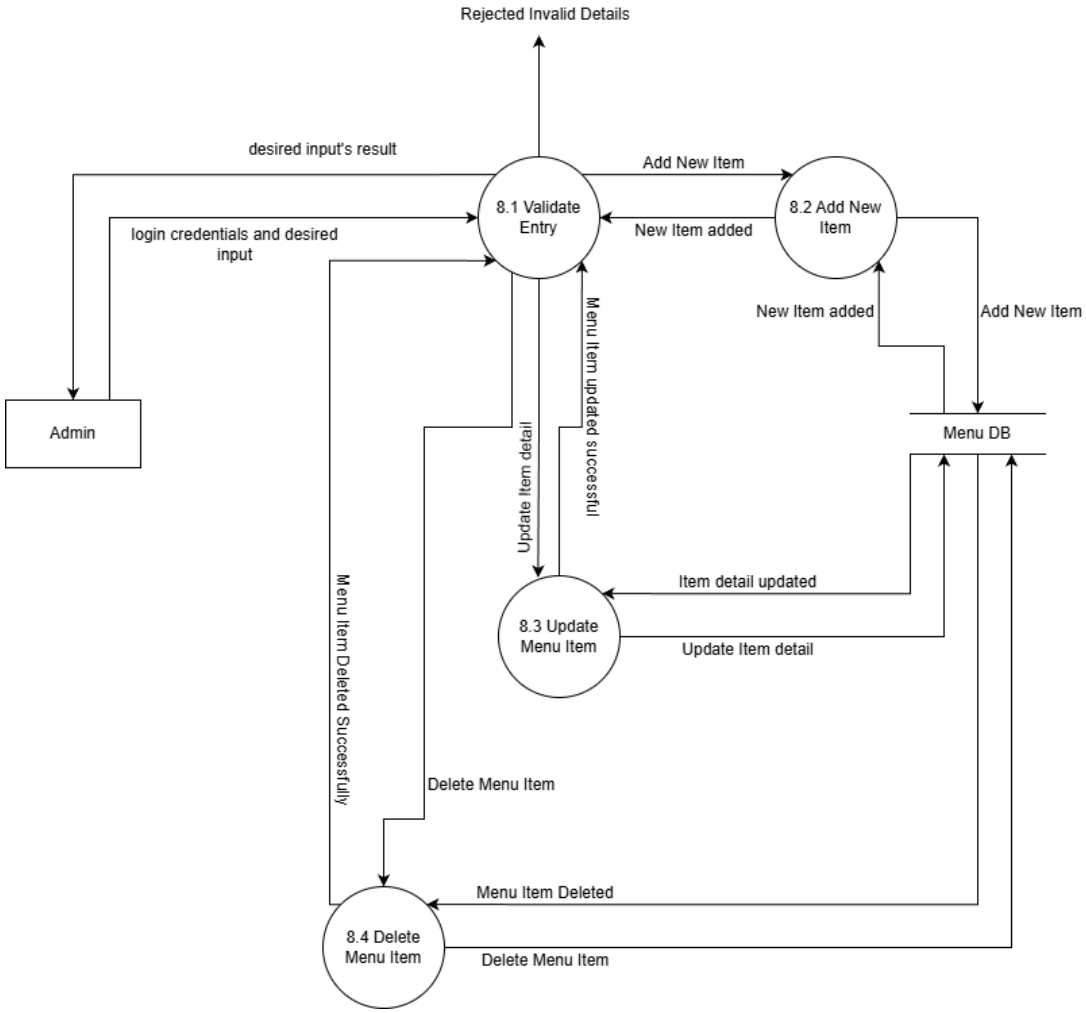


Figure 27 update menu DFD 1

3.5.3. Level 2 DFDs for the particular function

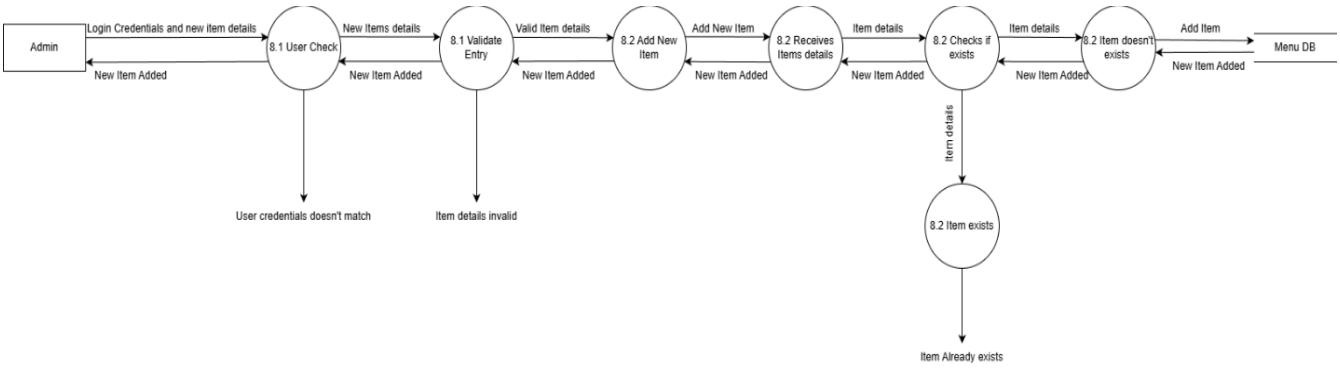


Figure 28 update menu DFD 2

3.5.4. Structure chart

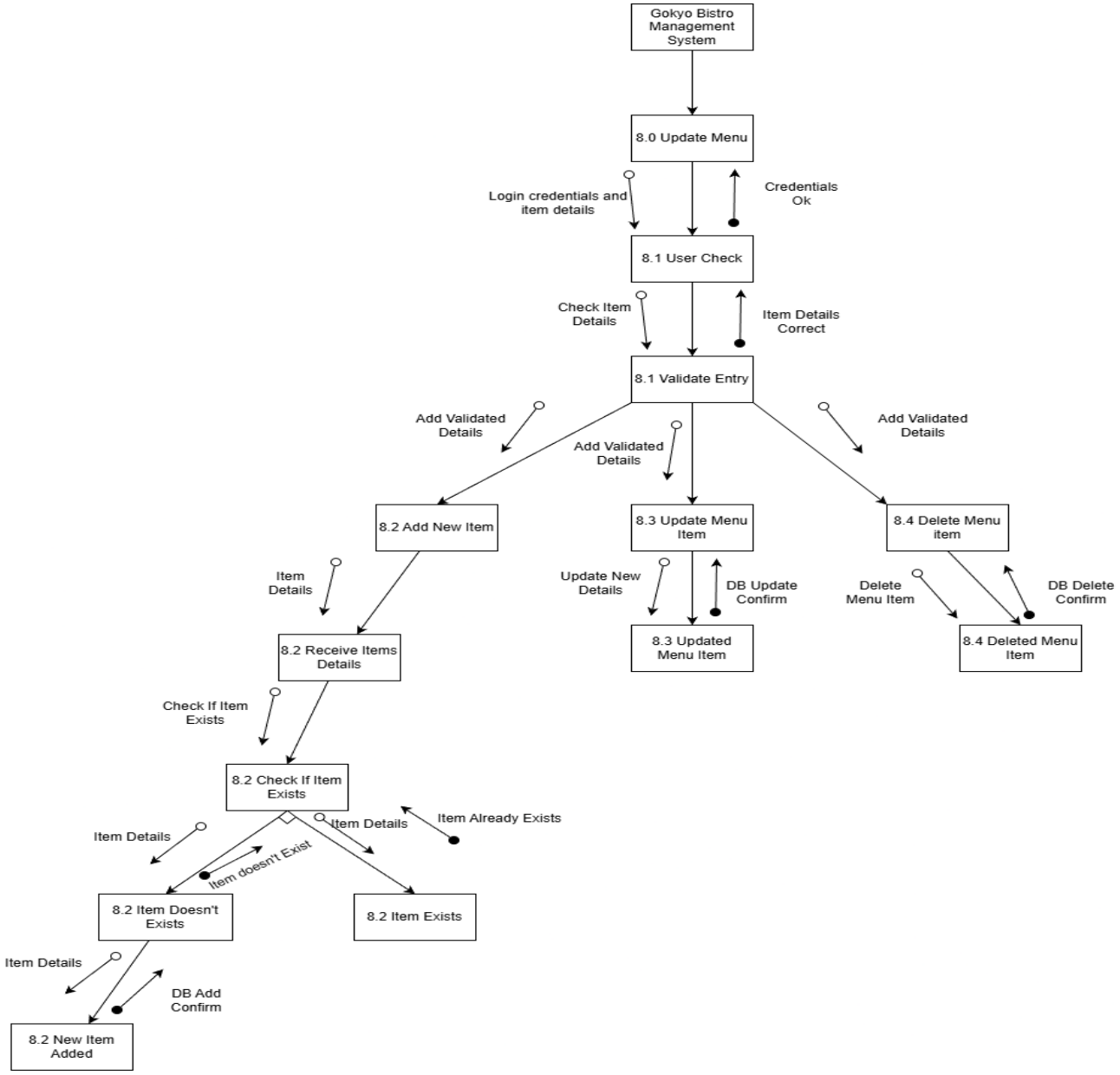


Figure 29 update menu structure chart

3.5.5. Module specifications

MODULE NAME: Update Menu

MODULE NUMBER: 8.0

PURPOSE: To serve as control module that lets user authentication, data validation, and do menu modification tasks.

PSEUDOCODE:

```
RECEIVE Login_Credentials and Action_Request (Add/Update/Delete).
CALL User_Check with credentials.
IF Credentials_Ok is True THEN:
    CALL Validate_Entry with raw item data.
    IF Item_Details_Correct is True THEN:
        IF Action is "Add" THEN CALL Add_New_Item.
        IF Action is "Update" THEN CALL Update_Menu_Item.
        IF Action is "Delete" THEN CALL Delete_Menu_Item.
    ELSE: RETURN "Rejected Invalid Details" to Admin.
ELSE: RETURN "Access Denied".
```

INPUT PARAMETERS:

- Admin_Id
- Password
- Item_Details, Action_Type

OUTPUT PARAMETERS:

- Final_Status_Message

GLOBAL VARIABLES: None

LOCAL VARIABLES:

- Credentials_Ok
- Item_Details_Correct

CALLS:

- User_Check
- Validate_Entry
- Add_New_Item
- Update_Menu_Item
- Delete_Menu_Item

CALLED BY: Gokyo_Bistro_Main_System

MODULE NAME: User Check

MODULE NUMBER: 8.1

PURPOSE: To authenticate the admin's login credentials before allowing menu modifications.

PSEUDOCODE:

RECEIVE Login_Credentials.

QUERY Admin_DB for matching Username and Password.

IF credentials match **THEN:**

SET Credentials_Ok = **True**.

ELSE:

SET Credentials_Ok = **False**.

RETURN Credentials_Ok

INPUT PARAMETERS:

- Username
- Password

OUTPUT PARAMETERS:

- Credentials_Ok (Either True or False)

GLOBAL VARIABLES:

- User_DB

LOCAL VARIABLES:

- Input_Credentials
- Price_Value

CALLS:

- Database_Auth_Library

CALLED BY: Update Menu

MODULE NAME: Validate Entry

MODULE NUMBER: 8.1

PURPOSE: To verify that the new menu item data follows business rules (e.g., non-empty name, positive price).

PSEUDOCODE:

```
RECEIVE Item_Name, Price, Category.

IF Item_Name is empty OR Price <= 0 THEN:

        SET Item_Details_Correct = False.

ELSE:

        SET Item_Details_Correct = True.

RETURN Item_Details_Correct.
```

INPUT PARAMETERS:

- Item_Details

OUTPUT PARAMETERS:

- Item_Details_Correct (Either True or False)

GLOBAL VARIABLES:

- Menu Data

LOCAL VARIABLES:

- Item_name_Length
- Price_Value

CALLS:

- String_Library
- Numeric_Validator

CALLED BY: Update Menu

MODULE NAME: Add New Item

MODULE NUMBER: 8.2

PURPOSE: To manage the sequence of adding a new item after it has been validated.

PSEUDOCODE:

```
CALL Receive_Items_Details.

CALL Check_If_Item_Exists.

IF Item_Exists is False THEN:

        CALL New_Item_Added.

ELSE:

        RETURN "Item Already Exists" to Admin.
```

INPUT PARAMETERS:

- Receive_Items_Details
- Check_If_Item_Exists

- New_Item_Added

OUTPUT PARAMETERS:

- Delete_Status

GLOBAL VARIABLES: None

LOCAL VARIABLES:

- Process_Step
- Exist_Flag

CALLS:

- Receive_Items_Details
- Check_If_Item_Exists
- New_Item_Added

CALLED BY: Update Menu

MODULE NAME: Receive Item Details

MODULE NUMBER: 8.2

PURPOSE: To capture and package the validated menu item attributes.

PSEUDOCODE:

RECEIVE Validated_Name.

RECEIVE Validated_Price.

RECEIVE Validated_Category.

CREATE New_Item_Object.

SET New_Item_Object.Name = Validated_Name.

SET New_Item_Object.Price = Validated_Price.

SET New_Item_Object.Category = Validated_Category.

RETURN New_Item_Object.

INPUT PARAMETERS:

- Validated_Name
- Validated_Price
- Validated_Category

OUTPUT PARAMETERS:

- New_Item_Object

GLOBAL VARIABLES: None

LOCAL VARIABLES:

- New_Item_Object

CALLS: None

CALLED BY: Add New Item

MODULE NAME: Check If Item Exists

MODULE NUMBER: 8.2

PURPOSE: To prevent duplicate items in the Menu Database.

PSEUDOCODE:

RECEIVE Item_Name**.**

SEARCH Menu_DB **for** Item_Name**.**

IF Name is found **THEN:**

SET Exist_Flag **= True** (Item Exists branch)**.**

ELSE:

SET Exist_Flag **= False** (Item Doesn't Exist branch)**.**

RETURN Exist_Flag.

INPUT PARAMETERS:

- Item_Name

OUTPUT PARAMETERS:

- Exist_Flag

GLOBAL VARIABLES:

- Menu DB

LOCAL VARIABLES:

- Search_Query
- Record_Count

CALLS:

- SQL_Search_Library

CALLED BY: Add New Item

MODULE NAME: New Item Added

MODULE NUMBER: 8.2

PURPOSE: To perform the final database, write operation.

PSEUDOCODE:

RECEIVE Validated_Item_Details.

INSERT new record into Menu_DB.

SET DB_Add_Confirm = **True.**

RETURN "New Item added Successfully" to Admin.

INPUT PARAMETERS:

- Validate_Item_Details

OUTPUT PARAMETERS:

- Item Added

GLOBAL VARIABLES:

- Menu DB
- System Log

LOCAL VARIABLES:

- Next_Available_id
- Write_Status

CALLS:

- ID_Generation_Service
- DB_Commit_Function

CALLED BY: Add New Item

4. Conclusion

In this coursework, we analyzed and designed the Gokyo Bistro Online Management System to address real problems faced by a growing restaurant. We observed that manual handling of table reservations, orders and billing often leads to inefficiency and customer dissatisfaction, especially during busy periods.

By applying structured software engineering, we were able to clearly understand the system requirements and represent them using the business case, SRS, DFDs, Data Dictionary and Process specifications. These tools helped us break down the system logically and ensured that all user and business needs were properly captured.

We also carried out a business and financial analysis, which showed that the proposed system is both practical and economically feasible in the local context. Features such as advance table booking, controlled cancellations and member-based home delivery support better revenue management and improved customer experience.

In a nutshell, this coursework helped us contact software engineering theory with a real-world scenario and provided us with a strong foundation for future system development and implementation.

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